

What is Excel?

Microsoft Excel is a spreadsheet program developed by Microsoft. There are other spreadsheet software like Google sheets, Zoho Office suit and LibreOffice, But Excel is the highest used among

Why do we need Excel?

Excel remains essential because **most of the businesses globally** use it for data tasks due to its versatility, ease of use, and widespread familiarity. It handles everything from simple **data entry to complex financial modelling** and integrates with other tools like Power BI. Its automation capabilities and adaptability make it invaluable for both **small and large-scale data needs.**

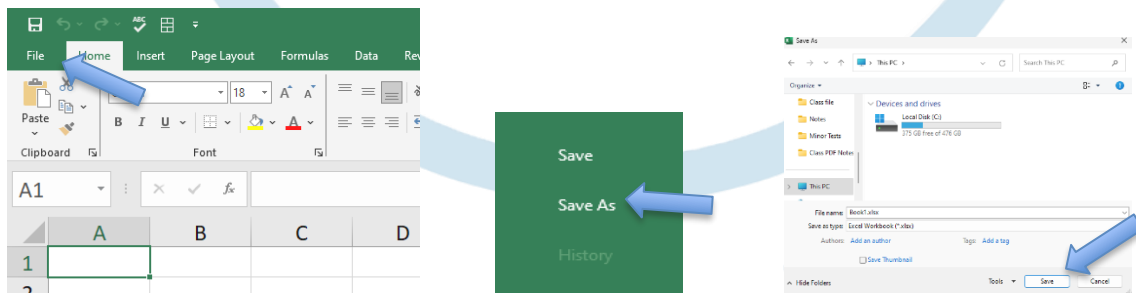
Features of Excel

Here are the features of Excel with a short explanation:

- **Data Entry:** Excel allows users to input data manually into cells or import it from external sources like CSV files, databases, or web pages.
- **Formatting:** Excel offers a variety of formatting options such as font styles, colors, cell borders, and number formats (currency, percentage, date), allowing users to customize the appearance of their data for better readability
- **Calculations:** With built-in formulas and functions (like SUM, AVERAGE, IF), Excel enables quick calculations, data manipulation, and complex computations.

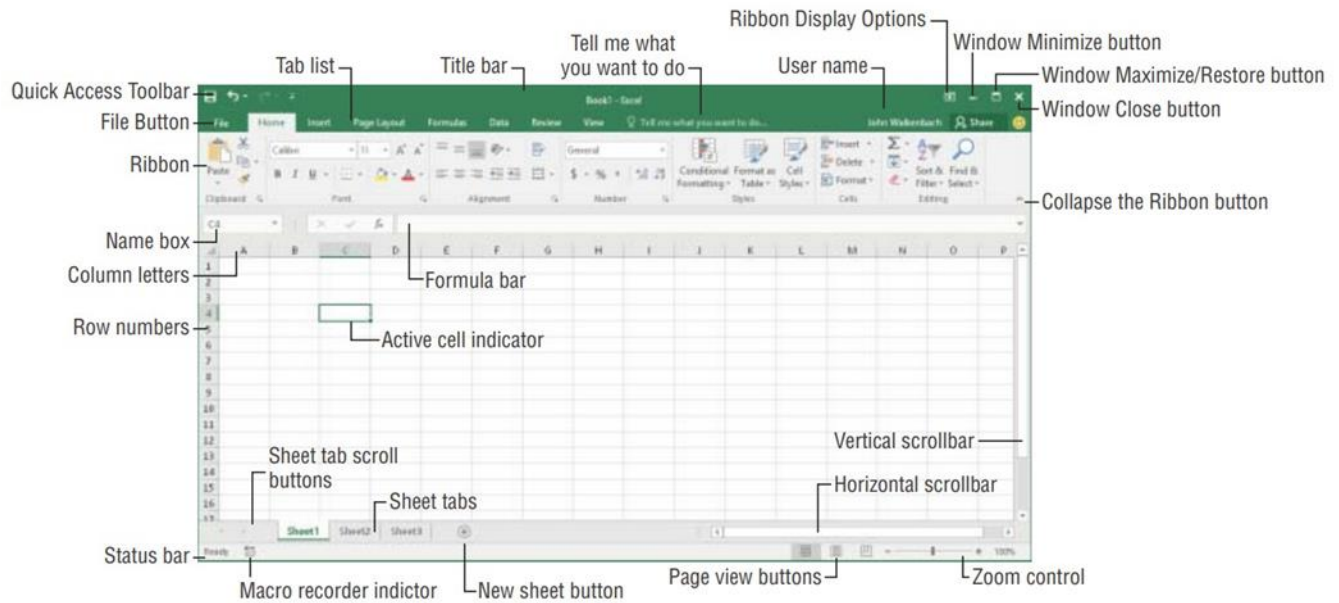
- **Chart Creation:** Excel provides several chart types (bar, pie, line, etc.) to visually represent data trends and comparisons, making data analysis easier to understand.
- **Data Analysis:** Tools like What-If Analysis filters and sorting help users model different scenarios and explore the effects of data changes.
- **PivotTables:** PivotTables are powerful tools for summarizing large datasets. They allow users to group, filter, and aggregate data dynamically, making it easy to extract insights from complex information.
- **Data Validation:** This feature restricts the type of data or value range that can be entered into a cell, ensuring accuracy and consistency by allowing only valid inputs (e.g., limiting inputs to numbers within a specific range).

Saving an Excel file

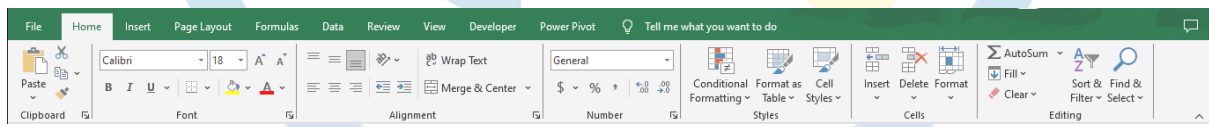


1. Go to file Tab
2. Select save as
3. Browse and select the location to save and save the file

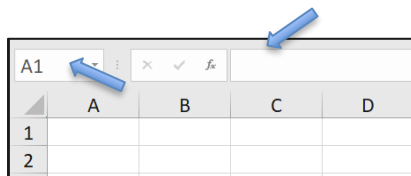
Excel Interface



Tabs: Different sections on the Ribbon (e.g., Home, Insert) that group related commands.

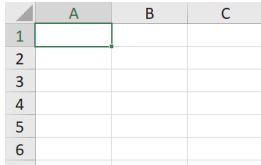


Ribbon: The toolbar at the top of the window, organized into tabs (like Home, Insert, and Formulas) that provide access to Excel's tools and commands.



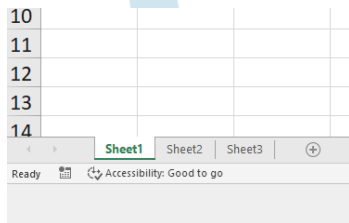
- **Name Box:** Displays the cell address or name of the currently selected cell.

- **Cell Address (Reference):** The unique location of a cell, identified by its column letter and row number (e.g., A1).
- **Formula Bar:** The area where you can enter or view data and formulas in the selected cell.

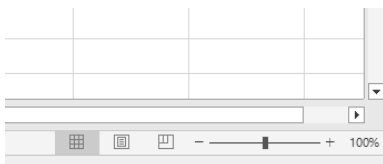


	A	B	C
1			
2			
3			
4			
5			
6			

- **Rows:** Horizontal lines in a worksheet, labelled with numbers.
- **Columns:** Vertical lines in a worksheet, labelled with letters.
- **Gridlines:** The lines that separate cells on a worksheet, making it easier to read data.



- **Sheet Tabs:** Located at the bottom of the workbook, these tabs allow you to navigate between different worksheets.



- **Status Bar:** Located at the bottom of the window, it provides information about the current operation or selection (e.g., sum of selected cells).
- **Scroll Bars:** Allow you to move horizontally or vertically through the worksheet.

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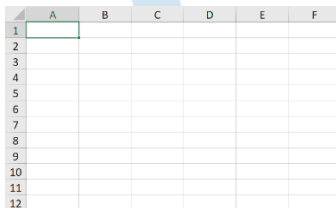
Important terms in Excel

Workbook:



A workbook in Excel is the file in which you store and manage your data. It contains one or more worksheets (also known as spreadsheets). Each workbook can hold multiple sheets. In technical terms **a single Excel file is called a Workbook.**

Worksheet:



A worksheet in Excel is a single page or tab within a workbook where you enter, organize, and analyse data. Each worksheet consists of a grid of cells arranged in rows and columns. There can be as many **till Excel 2003 excel support only 255 worksheets per book** but after that its **now unlimited.**

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Table:

	A	B	C	D
1	Product	Qtr 1	Qtr 2	Grand Total
2	Chocolade	\$744.60	\$162.56	\$907.16
3	Gummibarchen	\$5,079.60	\$1,249.20	\$6,328.80
4	Scottish Longbreads	\$1,267.50	\$1,062.50	\$2,330.00
5	Sir Rodney's Scones	\$1,418.00	\$756.00	\$2,174.00
6	Tarte au sucre	\$4,728.00	\$4,547.92	\$9,275.92
7	Chocolate Biscuits	\$943.89	\$349.60	\$1,293.49
8	Total	\$14,181.59	\$8,127.78	\$22,309.37

A Table is a collection of Rows and Columns. The first row being Header which contains name of each column.

Row:

	A	B	C	D	E
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

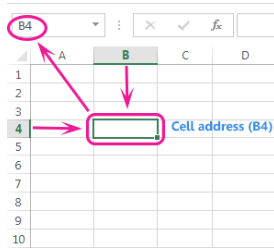
A horizontal line of cells in a worksheet, labelled with numbers (1, 2, 3, etc.). Each row contains data for a particular record or entry. There are total of **1,048,576 Rows in excel.**

Column:

	A	B	C	D	E
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					

A vertical line of cells in a worksheet, labelled with letters (A, B, C, etc.). Each column holds data for a specific category or field. There are total of **16,384 Columns in Excel.**

Cell:



A cell in Excel is a single block in a worksheet where you can enter data. Each cell is the intersection of a row and a column. Cells are identified by their unique cell Address, which is a combination of the column letter and row number (e.g., A1, B5).

Navigation in Excel Worksheet

Keyboard Shortcuts:

Arrow Keys	Move one cell at a time (Up, Down, Left, Right).
Ctrl + Arrow Keys	Jump to the edge of a data range or worksheet.
Tab	Move to the next cell on the right.
Shift + Tab	Move to the previous cell on the left.
Enter	Move down one cell.
Shift + Enter	Move up one cell.
Ctrl + Home	Go to cell A1 (start of the worksheet).
Ctrl + End	Go to the last used cell.
Ctrl + Page Up / Page Down	Move between worksheet tabs.
Mouse	Click on any cell to navigate directly.
Scroll Bar	Use the horizontal or vertical scroll bars to move through the worksheet.

Essential Keyboard Shortcuts:

Undo (Ctrl + Z)	Revert the last action or set of actions.
Redo (Ctrl + Y)	Repeat the last undone action.
Copy (Ctrl + C)	Copy the selected data.
Paste (Ctrl + V)	Paste the copied data to the selected location.
Bold (Ctrl + B)	Make selected text or numbers bold.
Italics (Ctrl + I)	Apply italic formatting to selected text or numbers.
Underline (Ctrl + U)	Underline the selected text or numbers.
Moving Data (Select and Drag)	Click and hold the edge of a selected cell or range of cells, then drag to a new location to move the data.

That's Great!

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Managing Data with Excel

What is Data?

EmpID	Name
101	Ramesh

Data refers to raw facts, figures, or information that can be recorded, measured, or analyzed. In Excel, data can be numbers, text, dates. For example, sales figures, names, or dates of transactions are all data.

What is a Database?

A database is an organized **collection of Tables**, typically stored and accessed electronically. In Excel, a database is often represented as a table or worksheet that stores large amounts of data.

What is a Row?

	A	B	C	D
1	Product	Qtr 1	Qtr 2	Grand Total
2	Chocolade	\$744.60	\$162.56	\$907.16
3	Gummibarchen	\$5,079.60	\$1,249.20	\$6,328.80
4	Scottish Longbreads	\$1,267.50	\$1,062.50	\$2,330.00
5	Sir Rodney's Scones	\$1,418.00	\$756.00	\$2,174.00
6	Tarte au sucre	\$4,728.00	\$4,547.92	\$9,275.92
7	Chocolate Biscuits	\$943.89	\$349.60	\$1,293.49
8	Total	\$14,181.59	\$8,127.78	\$22,309.37

A row in Excel is a **horizontal series of cells**. A single row contains all the details for a specific entity or record.

What is a Column?

	A	B	C	D
1	Product	Qtr 1	Qtr 2	Grand Total
2	Chocolade	\$744.60	\$162.56	\$907.16
3	Gummibarchen	\$5,079.60	\$1,249.20	\$6,328.80
4	Scottish Longbreads	\$1,267.50	\$1,062.50	\$2,330.00
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6	Tarte au sucre	\$4,728.00	\$4,547.92	\$9,275.92
7	Chocolate Biscuits	\$943.89	\$349.60	\$1,293.49
8	Total	\$14,181.59	\$8,127.78	\$22,309.37

A column is a **vertical series of cells** in a worksheet, which contains a specific type of data for all entity, like customer names, product prices, or Age.

What is a Table?

	A	B	C	D
1	Product	Qtr 1	Qtr 2	Grand Total
2	Chocolade	\$744.60	\$162.56	\$907.16
3	Gummibarchen	\$5,079.60	\$1,249.20	\$6,328.80
4	Scottish Longbreads	\$1,267.50	\$1,062.50	\$2,330.00
5	Sir Rodney's Scones	\$1,418.00	\$756.00	\$2,174.00
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7	Chocolate Biscuits	\$943.89	\$349.60	\$1,293.49
8	Total	\$14,181.59	\$8,127.78	\$22,309.37

A table in Excel is a **Collection of Rows and Columns** in which the first row has headers. Tables make it easier to analyze, sort, and filter data.

What are Table Headers?

	A	B	C	D
1	Product	Qtr 1	Qtr 2	Grand Total
2	Chocolade	\$744.60	\$162.56	\$907.16
3	Gummibarchen	\$5,079.60	\$1,249.20	\$6,328.80
4	Scottish Longbreads	\$1,267.50	\$1,062.50	\$2,330.00
5	Sir Rodney's Scones	\$1,418.00	\$756.00	\$2,174.00
6	Tarte au sucre	\$4,728.00	\$4,547.92	\$9,275.92
7	Chocolate Biscuits	\$943.89	\$349.60	\$1,293.49
8	Total	\$14,181.59	\$8,127.78	\$22,309.37

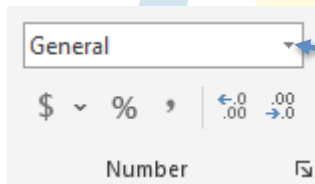
Table headers are **the labels at the top of each column** in an Excel table. They define what type of information each column contains (e.g., "Name," "Date," "Price"). Headers make it easier to identify and understand the data in each column.

Datatypes in Excel

What are Datatypes?

Data types refer to the classification or category different kinds of data. Data types determine how data is stored, displayed, and manipulated such as numbers, text, dates, etc. is treated when performing calculations or applying formatting.

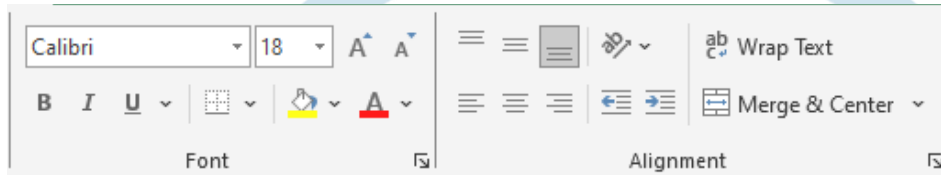
Different Datatypes in Excel



- **General** Default format for any data. Displays numbers as entered without special formatting.
- **Number** Displays numbers with options for decimals and negative values.
- **Currency** Formats numbers as monetary values, with currency symbols (e.g., \$) and decimals.
- **Date** Formats cells to display dates in various formats (e.g., dd/mm/yyyy).
- **Time** Formats cells to display time in formats like hh:mm AM/PM.
- **Percentage** Multiplies the number by 100 and adds a percentage sign (e.g., 0.5 becomes 50%).
- **Scientific** Displays numbers in scientific notation (e.g., 1.23E+04 for 12300).
- **Fraction** Displays numbers as fractions (e.g., 0.75 becomes 3/4).

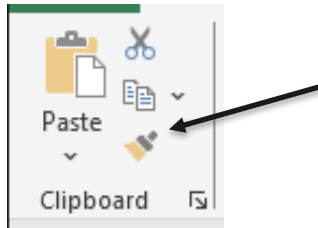
- **Text** Treats data as text, even if it's a number, so no calculations are performed.

Data Manipulation & Formatting



- **Data Alignment** Controls how data is aligned in a cell (left, centre, right, top, middle, bottom).
- **Font Color, Font Size** Allows you to change the color and size of the text within a cell for better readability or emphasis.
- **Cell Color** Lets you fill a cell with a background color to highlight or differentiate data.
- **Row Height** Adjusts the height of a row to fit content or for better visual presentation.
- **Column Width** Changes the width of columns to accommodate longer text or data.
- **Merge** Combines multiple adjacent cells into a single larger cell.
- **Wrap Text** Ensures that long text is displayed on multiple lines within a single cell without overflowing into other cells.

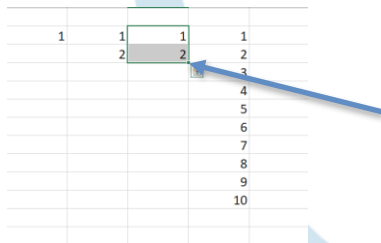
Format painter



The Format Painter is a tool that allows you to quickly copy the formatting from one cell (or range of cells) and apply it to another. This includes formatting like font style, font size, cell color, borders, and alignment.

Auto Fill

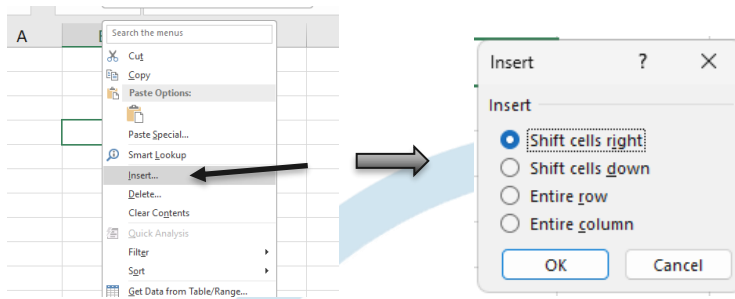
Auto Fill is a feature that automatically fills data in adjacent cells based on the pattern or sequence in the selected cells. It works for numbers, dates, text, or formulas.



How it Works:

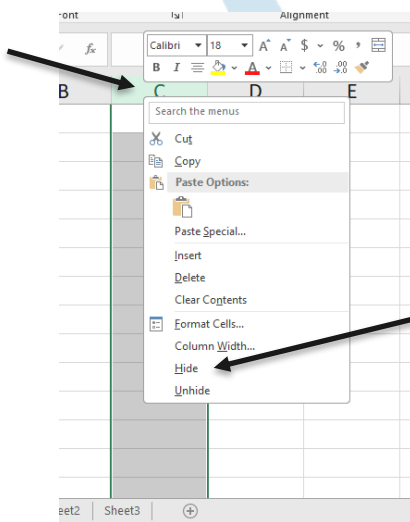
- Enter data in a cell (e.g., a number, date, or formula).
- Select the cell, then drag the fill handle (a small square at the bottom-right corner of the cell) over adjacent cells.
- Excel automatically fills the selected cells based on the pattern (e.g., incrementing numbers, continuing a date sequence, copying text, Weeks names, Month Names etc.)

Inserting/Deleting Rows and Columns

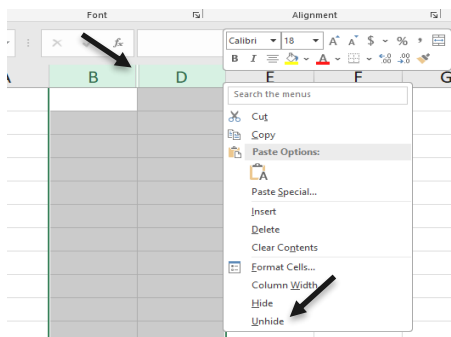


- To insert a row or column in Excel right click on any cell and select insert than select any option or you can directly select a column or row and press ctrl+Shift+'+'
- **Note:** To insert multiple rows. Select the number of rows you want to add. right-click within the selected area. click on 'Insert' from the options provided.

Hiding/Unhiding Rows and Columns



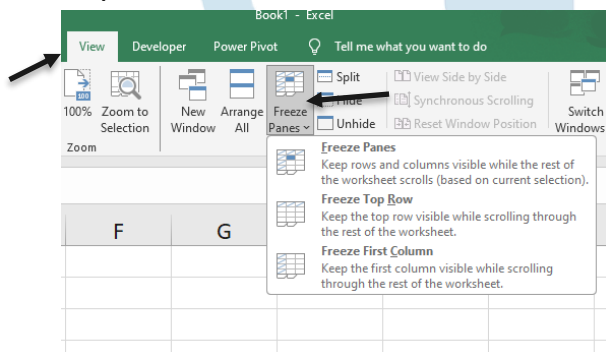
- First right click on any row or column and select it by right clicking on it then select Hide.



- To check which column is hidden look at the column B and D there is no C because it is hidden to unhide it selects column from left to right then right click on it then select Unhide.

Freeze Panes

Freeze Panes allows you to keep specific rows or columns visible while scrolling through the rest of the worksheet. This is useful for keeping headers or important information in view.

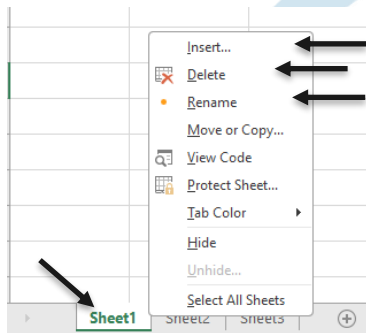


To use the Freeze Pane, go to the View tab and select Freeze pane.

- **Freeze Top Row:** Keeps the first row visible when you scroll down. Useful for keeping headers in view.
- **Freeze First Column:** Keeps the first column visible when you scroll sideways. Helpful for identifying data categories.

- **Freeze Both Rows and Columns:** Allows you to freeze both specific rows and columns (e.g., freezing the top row and the first column).

Renaming, Moving, Copying and Deleting Sheets



- **Move** You can drag and drop the sheets to Move then
- **Rename** To rename right click and select rename
- **Copy** to copy a sheet right click and select copy
- **Delete** to delete a sheet right click and select delete

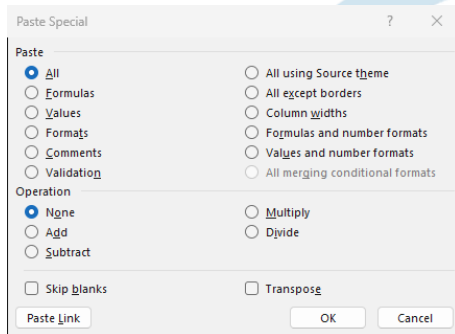
Copy & Paste Data

Original Data	EmpID	Name			
	101	Ramesh			
Paste Standard	EmpID	Name	Paste Transpose	EmpID	101
	101	Ramesh		Name	Ramesh
Paste values	EmpID	Name	Paste Format		
	101	Ramesh			

Paste (Ctrl + V)	Standard paste.
Paste Values	Pastes only the values (no formulas or formatting).
Paste Formulas	Pastes just the formulas.

Paste Formatting	Copies only the formatting.
Paste Transpose	Swaps rows and columns.

Paste Special



Excel's Paste Special offers a wide range of options to paste only specific elements of the copied cells or perform a mathematical operation with the copied data.

- For example, you can copy formula-driven data and paste only the calculated values in the same or different cells.
- You can copy the width of one column and apply it to all other columns in your data
- You can transpose the copied range, i.e. convert rows to columns and vice versa.

Datatypes

Data types refer to the kinds of values or information that can be stored in a cell. Each data type defines the nature of the data and how it should be processed or displayed.

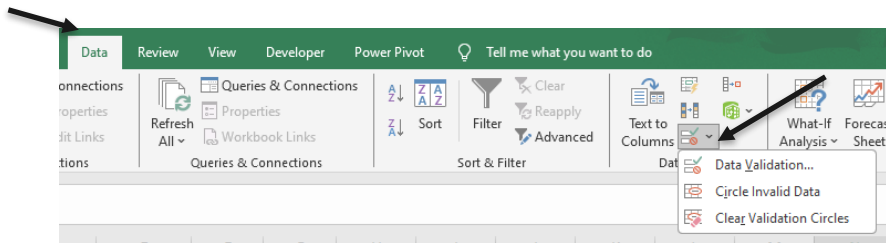
Common Data Types in Excel:

Text (String)	Any combination of letters, numbers, or symbols treated as text. It can include names, labels, or any non-numeric information. Example "John Doe", "Invoice #12345"
Numbers (Numeric)	Pure numeric data that can be used in calculations, such as integers, decimals, or negative numbers. Example 25, -120, 3.14
Date and Time	Special data types for representing dates and times. Excel treats dates as serial numbers for calculations. Example 14-Sep-2024, 12
Currency	A numeric data type that specifically represents monetary values, formatted with currency symbols. Example \$100.50, ₹1200
Percentage	Numeric data presented as a percentage. It is internally stored as a decimal, but shown as a percentage. Example 50% (stored as 0.50)
Boolean (Logical)	Represents logical values such as TRUE or FALSE, used in logical operations or formulas. Example TRUE, FALSE

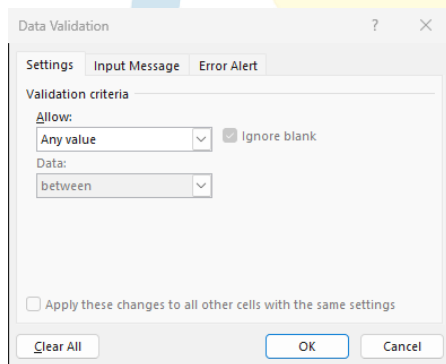
Why Understanding Data Types is Important in Excel:

- **Accurate Calculations:** Correct data types ensure that Excel can perform calculations and analyses without errors.
- **Formatting:** Properly understanding data types helps apply the correct formats (e.g., currency, percentage) to make the data presentable
- **Error Prevention:** Using the wrong data type can cause errors in formulas or misinterpretations of results.
- **Efficient Data Handling:** Knowing the data types helps in applying functions and formulas correctly.

Data validation Techniques



Data validation in Excel is a feature that allows you to control the type of data entered into a cell. It helps ensure that the data is accurate, consistent, and meets certain criteria. You can find the Data validation option in the **Data tab** - > **Data validation**.



As you click on a cell or range and select Data validation it gives you options to select criteria.

Any Value	No restrictions.
Whole Number	Only integers are allowed.
Decimal	Numbers with decimal points.
List	Dropdown list of predefined options.
Date	Dates within a specified range.

Time	Times within a specified range.
Text Length	Limits the number of characters.
Custom	Use a custom formula to define validation criteria.



Basic Operations in Excel

Operators in Excel

An operator in Excel is a symbol or function that specifies the type of calculation or comparison to perform on operands or elements in a formula.

Types of Operators

1. Arithmetic operators

Arithmetic operators in Excel are used to perform basic mathematical calculations.

Arithmetic operator	Meaning	Example
+	Addition	=3+3
-	Subtraction Negation	=3-1
*	Multiplication	=3*3
/	Division	=3/3
%	Percent	=20%
^	Exponentiation	=2^3

2. Comparison operator

Comparison operators in Excel are used to compare values and determine their relationships. They return a Boolean result (TRUE or FALSE) based on the comparison.

Comparison operator	Meaning	Example
=	Equal to	=A1=B1
>	Greater than	=A1>B1
<	Less than	=A1<B1
>=	Greater than or equal to	=A1>=B1
<=	Less than or equal to	=A1<=B1
<>	Not equal to	=A1<>B1

Order of Operations

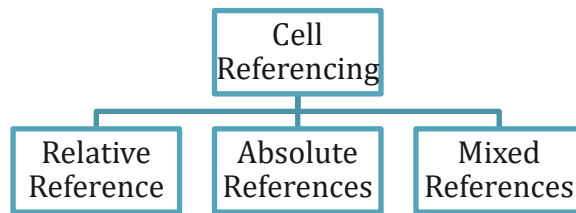
The BODMAS rule guides Excel in determining the Order of Operations, specifying the sequence for accurate calculations.

$$\begin{aligned}
 &36 \div 6 \times 3 + 2^2 - (3 + 5) \\
 &= 36 \div 6 \times 3 + 2^2 - \underline{8} \quad \longleftrightarrow \text{Brackets: } (3 + 5) \\
 &= 36 \div 6 \times 3 + \underline{4} - 8 \quad \longleftrightarrow \text{Order of Powers: } 2^2 \\
 &= \underline{6} \times 3 + 4 - 8 \quad \longleftrightarrow \text{Division: } 36 \div 6 \\
 &= \underline{18} + 4 - 8 \quad \longleftrightarrow \text{Multiplication: } 6 \times 3 \\
 &= \underline{22} - 8 \quad \longleftrightarrow \text{Addition: } 18 + 4 \\
 &= 14 \quad \longleftrightarrow \text{Subtraction: } 22 - 8
 \end{aligned}$$

1. Bracket
2. Operator (Exponents)
3. Division
4. Multiplication
5. Addition
6. Subtraction

Cell referencing

Cell Referencing in Excel refers to the method of **identifying a cell's location within a worksheet** using its rows (1,2,3) and columns (A,B,C) coordinates. This allows you to use cell values in formulas and functions, and to create dynamic calculations and data references.



Excel's Cell Reference Types	
Relative	A1
Absolute	\$A\$1
Mixed	\$A1 or A\$1

Types of Cell referencing

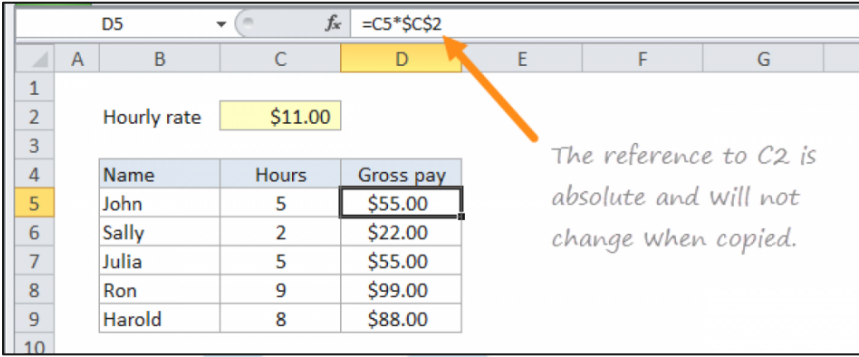
Relative References:

COUNTIF					
					=B2*C2
	A	B	C	D	E
1	Product	Quantity	Price	Amount	
2	bread	2	1.5	=B2*C2	
3	butter	1	1.2		
4	cheese	3	2		
5	ham	3	1.8		
6					

Adjust automatically when copied to another cell (e.g., B2 and C2). If you copy a formula with a relative reference from one cell to another, the reference adjusts based on its new position.

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Absolute References:



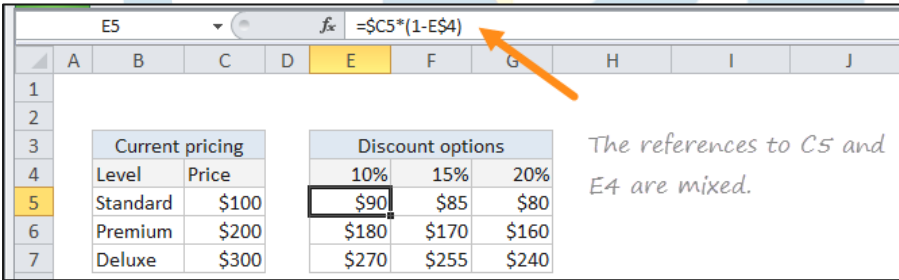
The screenshot shows an Excel spreadsheet. The formula bar at the top displays the formula `=C5*$C$2` for cell D5. An orange arrow points from the formula bar to cell D5. The spreadsheet contains a table for calculating gross pay based on hourly rate and hours worked. The formula in D5 is an absolute reference, meaning it will remain constant if the formula is copied to other cells.

	A	B	C	D	E	F	G
1							
2		Hourly rate	\$11.00				
3							
4		Name	Hours	Gross pay			
5		John	5	\$55.00			
6		Sally	2	\$22.00			
7		Julia	5	\$55.00			
8		Ron	9	\$99.00			
9		Harold	8	\$88.00			
10							

The reference to C2 is absolute and will not change when copied.

It Remain constant, no matter where the formula is copied (e.g., `$C$2`). Using absolute references ensures that the specific cell reference remains unchanged when the formula is moved or copied.

Mixed References:



The screenshot shows an Excel spreadsheet. The formula bar at the top displays the formula `=C5*(1-E$4)` for cell E5. An orange arrow points from the formula bar to cell E5. The spreadsheet contains two tables: 'Current pricing' and 'Discount options'. The formula in E5 is a mixed reference, meaning it will adjust based on the cell's position in some ways but remain constant in others.

	A	B	C	D	E	F	G	H	I	J
1										
2										
3		Current pricing			Discount options					
4		Level	Price		10%	15%	20%			
5		Standard	\$100		\$90	\$85	\$80			
6		Premium	\$200		\$180	\$170	\$160			
7		Deluxe	\$300		\$270	\$255	\$240			

The references to C5 and E4 are mixed.

Combine relative and absolute references (e.g., `$C5` and `E$4`). One part of the reference is fixed, while the other part adjusts based on the cell's position.

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Introduction to Excel Functions

What is Function?

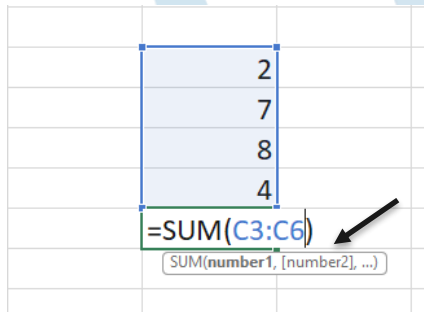
Function is a predefined formula that performs specific calculations or operations on data. Functions take arguments (input values), execute a predefined operation, and return a result.

Syntax of a function?

Syntax refers to the specific **structure and format you must use** when entering a function or formula.

=FUNCTION_NAME(argument1, argument2, ...)
=SUM(A1:A10)

What are Arguments?



Arguments are the inputs or values that you provide to the function so it can perform its task.

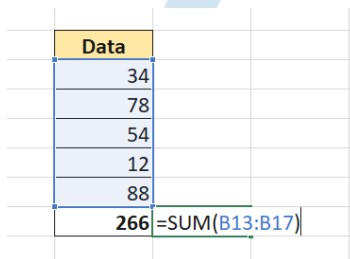
Types of Functions

- Numerical Functions
 - Aggregate Functions
 - Non-Aggregate Functions

- Text Functions
- Date & Time Functions
- Lookup Functions
- Logical Functions
- Statistical Functions

Numeric Functions – Aggregate

SUM:

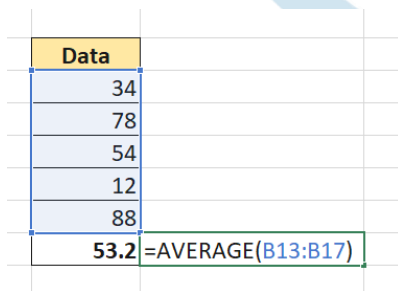


Data
34
78
54
12
88
266 =SUM(B13:B17)

=SUM(number1, [number2], ...)

SUM Function is use to find the total of a given array.

AVERAGE:



Data
34
78
54
12
88
53.2 =AVERAGE(B13:B17)

=AVERAGE(number1, [number2], ...)

Calculates the average (mean) of a range of numbers.

MAX:

Data
34
78
54
12
88
88 =MAX(B13:B17)

=MAX(number1, [number2], ...)

Returns the largest number in a range.

MIN:

Data
34
78
54
12
88
12 =MIN(B13:B17)

=MIN(number1, [number2], ...)

Returns the smallest number in a range.

COUNT:

Data
34
78
54
12
88
5 =COUNT(B13:B17)

Data
Milk
Chocolate
Table
Chair
Mouse
0 =COUNT(B27:B31)

=COUNT(value1, [value2], ...)

Counts the number of cells with only numeric values in a range.

COUNTA:

Data	
Milk	
Chocolate	
67	
Chair	
Mouse	
5	=COUNTA(B27:B31)

=COUNTA(value1, [value2], ...)

Counts the number of cells that are not empty it includes numbers and text also.

COUNTBLANK:

Data	
Milk	
Chair	
Mouse	
2	=COUNTBLANK(B27:B31)

=COUNTBLANK(range)

Count the number of blank cells in the data.

Non-Aggregate Functions

ROUND:

	A	B
35		
36	45.8954	
37	45.9	=ROUND(A36,1)

=ROUND(number, num_digits)

Round the number to specific decimal place.

POWER:

	A	B
39		
40	5	
41	125	=POWER(A40,3)

=POWER(number, power)

Raises a number to a specified power.

SQRT:

	A	B
1	225	
2	15	=SQRT(A1)

=SQRT(number)

Find the square root of a number.

Text Functions

TRIM

	A	B
1	Name	Trim
2	shelia Mahoney	shelia Mahoney
3	Jo Rivers	Jo Rivers
4	Sheryl Lowery	Sheryl Lowery
5	whitney hooper	whitney hooper
6	lindsey rice	=TRIM(A6)

Removes extra spaces from text from starting, end or middle.

=TRIM(text)

PROPER

	A	B
1	Name	PROPER
2	shelia MahOnEy	Shelia Mahoney
3	Jo RivERs	Jo Rivers
4	SHERYL LOWERY	Sheryl Lowery
5	whitney hoOPER	Whitney Hooper
6	LINDSEY RICE	=PROPER(A6)

Capitalizes the first letter of each word.

Syntax: =PROPER(text)

UPPER

	A	B
1	Name	UPPER
2	shelia Mahoney	SHELIA MAHONEY
3	Jo Rivers	JO RIVERS
4	sheryl Lowery	SHERYL LOWERY
5	whitney hooper	WHITNEY HOOPER
6	lindsey rice	=UPPER(A6)

Description: Converts all letters to uppercase.

Syntax: =UPPER(text)

LOWER

	A	B
1	Name	LOWER
2	SHELIA MAHONEY	shelia mahoney
3	JO RIVERS	jo rivers
4	SHERYL LOWERY	sheryl lowery
5	WHITNEY HOOPER	whitney hooper
6	LINDSEY RICE	=LOWER(A6)

Description: Converts all letters to lowercase.

Syntax: =LOWER(text)

CONCATENATE

	A	B	C
1	First Name	Last Name	Full Name
2	shelia	Mahoney	shelia Mahoney
3	Jo	Rivers	Jo Rivers
4	Sheryl	Lowery	Sheryl Lowery
5	whitney	hooper	whitney hooper
6	lindsey	rice	=CONCATENATE(A6," ",B6)

Description: Joins multiple text strings into one.

Syntax: =CONCATENATE(text1, [text2], ...)

LEFT

	A	B
1	Full Name	LEFT
2	shelia Mahoney	she
3	Jo Rivers	Jo
4	Sheryl Lowery	She
5	whitney hooper	whi
6	lindsey rice	=LEFT(A6,3)

Description: Extracts a specified number of characters from the beginning of a text string.

Syntax: =LEFT(text, [num_chars])

RIGHT

	A	B
1	Full Name	RIGHT
2	shelia Mahoney	oney
3	Jo Rivers	vers
4	Sheryl Lowery	wery
5	whitney hooper	oper
6	lindsey rice	=RIGHT(A6,4)

Description: Extracts a specified number of characters from the end of a text string. For example, extract last 4 letter from name.

Syntax: =RIGHT(text, [num_chars])

MID

	A	B
1	Full Name	MID
2	shelia Mahoney	el
3	Jo Rivers	R
4	Sheryl Lowery	er
5	whitney hooper	it
6	lindsey rice	=MID(A6,3,2)

Description: Extracts a specified number of characters from the middle of a text string.

Syntax: =MID(text, start_num, num_chars)

TEXT

Input	Output	Formula
100	\$100	=TEXT(B5,"\$#,##0")
0.85	85%	=TEXT(B6,"0%")
6:00 PM	18:00	=TEXT(B7,"hh:mm")
3.141592654	3.14	=TEXT(B8,"0.00")
25000000000	2.5E+10	=TEXT(B9,"0.0E+00")
17-Dec-2021	Friday, December 17	=TEXT(B10,"dddd, mmmm dd")
17-Dec-2021	December 17, 2021	=TEXT(B11,"mmm dd, yyyy")
17-Dec-2021	Fri, Dec 17	=TEXT(B12,"ddd, mmm dd")
15%	Profits are up 15%	="Profits are up "&TEXT(B13,"0%")
3-May-2021	Register by May 3!	="Register by "&TEXT(B14,"mmm d")&"!"

Description: Formats a number or date into text with a specified format.

Syntax: =TEXT(value, format_text)

List of TEXT Function formatting

General number	"0"
Fixed decimals	"0.00"
Thousands separator	"#,##0"
Percent	"0%"
Currency	"\$#,##0.00"

Scientific notation	"0.00E+00"
Day (1-31)	"d"
Day (01-31)	"dd"
Day (Sun-Sat)	"ddd"
Day (Sunday-Saturday)	"dddd"
Month (1-12)	"m"
Month (01-12)	"mm"
Month (Jan-Dec)	"mmm"
Month (January-December)	"mmmm"
Year (12)	"yy"
Year (2012)	"yyyy"
Date and time together	"dd-mm-yyyy hh:mm:ss"

Date Functions

NOW

	A	B
1	9/20/2024 13:39	=NOW()

=NOW()

Returns the current date and time.

TODAY

	A	B
1	9/20/2024	=TODAY()

=TODAY()

Returns the current date without the time.

YEAR

3	Date	Year	Formula
4	11/12/2017	2017	=YEAR(A4)
5	11/1/2015	2015	(YEAR(serial_number))
6	3/11/1984	1984	
7	12/11/2000	2000	
8	6/30/2012	2012	
9	1/25/2002	2002	

=YEAR(serial_number)

Extracts the year from a given date.

MONTH

3	Date	Month
4	11/12/2017	11
5	11/1/2015	11
6	3/11/1984	3
7	12/11/2000	12
8	6/30/2012	6
9	1/25/2002	=MONTH(A9)

Extracts the month from a given date as a number (1 for January, 2 for February, etc.).

Syntax: =MONTH(serial_number)

DAY

3	Date	Day
4	11/12/2017	12
5	11/1/2015	1
6	3/11/1984	11
7	12/11/2000	11
8	6/30/2012	30
9	1/25/2002	=DAY(A9)

Extracts the day of the month from a given date.

Syntax: =DAY(serial_number)

DATE

14		
15	2/21/2024	=DATE(2024,2,21)
16		

Creates a date based on the year, month, and day provided.

Syntax:=DATE(year, month, day)

WEEKDAY

3	Date	Weekday
4	11/12/2017	1
5	11/1/2015	1
6	3/11/1984	1
7	12/11/2000	2
8	6/30/2012	7
9	1/25/2002	=WEEKDAY(A9)

Returns the day of the week for a given date as a number. By default, 1 = Sunday, 2 = Monday, and so on.

Syntax:=WEEKDAY(serial_number, [return_type])

WEEKNUM

3	Date	Weeknum
4	11/12/2017	46
5	11/1/2015	45
6	3/11/1984	11
7	12/11/2000	51
8	6/30/2012	26
9	1/25/2002	=WEEKNUM(A9)

Returns the week number of a specific date in a year of 1 to 52 Week.

Syntax:=WEEKNUM(serial_number, [return_type])

DATEDIF

3	Date	Date 2nd	Datediff
4	6/22/1997	9/9/2014	17
5	4/27/2001	3/4/2019	17
6	12/19/2002	1/11/2021	18
7	8/25/1999	5/1/2010	10
8	6/13/1998	11/24/2019	21
9	2/4/2007	11/11/2013	=DATEDIF(A9,B9,"Y")
10			(DATEDIF)

Calculates the difference between two dates in years, months, or days. This function is undocumented in Excel.

Syntax:=DATEDIF(start_date, end_date, unit)

unit: The type of difference to return.

"Y"	Returns the number of complete years.
"M"	Returns the number of complete months.
"D"	Returns the number of days.
"MD"	Returns the difference in days, ignoring months and years.
"YM"	Returns the difference in months, ignoring days and years.
"YD"	Returns the difference in days, ignoring years.

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Advanced Excel Functions

Advance Logic Functions

IF:

The screenshot shows the Excel formula bar with the formula `=IF(B2>80, "Good", "Bad")`. Above the formula bar, labels indicate the parts of the function: `logical_test` for `B2>80`, `value_if_true` for `"Good"`, and `value_if_false` for `"Bad"`. Below the formula bar is a table with columns A, B, and C.

	A	B	C
1	Name	Score	Result
2	Aiden	88	Good
3	Ava	61	Bad
4	Emma	94	Good
5	Ethan	73	Bad
6	Liam	64	Bad
7	Lucas	83	Good

Returns one value if a condition is TRUE, and another if it is FALSE.

`=IF(A1 > 10, "Yes", "No")` → Returns "Yes" if A1 is greater than 10, otherwise "No."

AND:

The screenshot shows the Excel formula bar with the formula `=AND(A35,B35)`. Below the formula bar is a table with columns A, B, and C.

	A	B	C
34	Condition 1	Condition 2	AND
35	TRUE	TRUE	TRUE
36	TRUE	FALSE	FALSE
37	FALSE	TRUE	FALSE
38	FALSE	FALSE	FALSE

Returns TRUE if all conditions are TRUE otherwise, it returns FALSE.

`=AND(A1 > 10, B1 < 5)` → TRUE if both conditions are true.

OR:

C35 =OR(A35,B35)

	A	B	C
34	Condition 1	Condition 2	OR
35	TRUE	TRUE	TRUE
36	TRUE	FALSE	TRUE
37	FALSE	TRUE	TRUE
38	FALSE	FALSE	FALSE

Returns TRUE if any of the conditions are TRUE.

$=OR(A1 > 10, B1 < 5) \rightarrow TRUE$ if at least one condition is true.

NOT

B43		=NOT(A43)	
	A	B	
42	True/False	NOT	
43	TRUE	FALSE	
44	FALSE	TRUE	

Reverses the logical value; returns TRUE if the condition is FALSE, and vice versa.

$\text{=NOT}(A1 > 10) \rightarrow$ Returns TRUE if A1 is not greater than 10

NESTED IF

Formula Bar: `=IF(B13>=90,"A",IF(B13>=70,"B",IF(B13>=50,"C","D")))`

	A	B	C	D	E	F
11	Q1. Marksheet				Criteria	
12	Student Name	Marks	Grade		Grade	Marks
13	Nimisha	35	D		A	>=90
14	Nikhil	60	C		B	>=70
15	Nikita	92	A		C	>=50
16	Naresh	74	B		D	<50
17	Maneesh	85	B			
18	Mohit	48	D			
19	Naman	64	C			
20	Kartik	97	A			
21	Kaju	22	D			

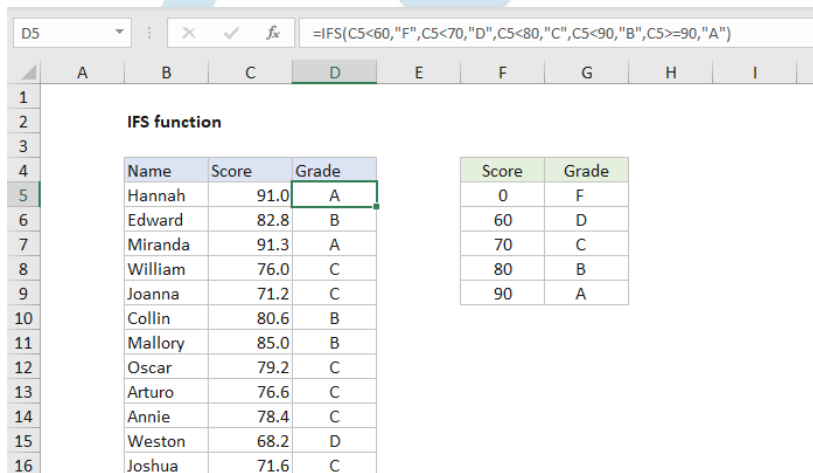
Grade the students on the bases on their mark's.

Grade the students on the bases on their mark's.

A nested IF is when multiple IF functions are used inside one another to test multiple conditions. It allows you to perform different actions based on several criteria just like in the above example of students.

Syntax: =IF(condition1, value_if_true1, IF(condition2, value_if_true2, value_if_false2))

IFs



The screenshot shows an Excel spreadsheet with the formula bar at the top displaying the IFS function: `=IFS(C5<60,"F",C5<70,"D",C5<80,"C",C5<90,"B",C5>=90,"A")`. The spreadsheet contains a table of student names, scores, and grades. The formula is applied to cell D5, which contains the grade 'A' for Hannah, whose score is 91.0.

Name	Score	Grade
Hannah	91.0	A
Edward	82.8	B
Miranda	91.3	A
William	76.0	C
Joanna	71.2	C
Collin	80.6	B
Mallory	85.0	B
Oscar	79.2	C
Arturo	76.6	C
Annie	78.4	C
Weston	68.2	D
Joshua	71.6	C

Score	Grade
0	F
60	D
70	C
80	B
90	A

The IFS function in Excel is a simpler alternative to nested IF statements. It evaluates multiple conditions and returns a result for the first TRUE condition.

Syntax: =IFS(condition1, value_if_true1, condition2, value_if_true2, ..., [condition_n, value_if_true_n])

Advanced TEXT Functions

REPLACE

B49					=REPLACE("A123B", 2, 3, "ABC")
		A		B	C
48		Old Text		Replaced Text	
49		A123B		AABCB	

The REPLACE function replaces part of a text string with another text string based on the position you specify.

Syntax:=REPLACE(old_text, start_num, num_chars, new_text)

FIND

B49					=FIND("a",A49)
		A		B	
48		Text		Find A	
49		Name			2

The FIND function locates the position of a specific character or substring within a text string. It is case-sensitive.

Syntax:=FIND(find_text, within_text, [start_num])

SEARCH

B49					=SEARCH("A",A49)
		A		B	
48		Text		Search	
49		Name			2

The SEARCH function is similar to FIND, but it is not case-sensitive.

Syntax: =SEARCH(find_text, within_text, [start_num])

Search function can also be used with Wild cards in order to search for missing letter.

? (Question Mark): Represents any single character.

Example: "A?C" matches "ABC", "ACC", etc.

*** (Asterisk):** Represents any number of characters (zero or more).

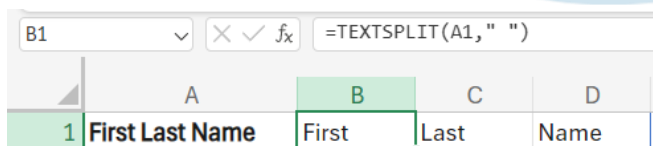
Example: "A*C" matches "ABC", "ABBC", "AC", etc.

~ (Tilde): Used to escape wildcards. If you want to search for the actual characters ?, *, or ~, use ~ before them.

Example: "A~*C" matches "A*C" (with an actual asterisk), rather than using * as a wildcard.

Key Differences: FIND is case-sensitive whereas SEARCH is case-insensitive

TEXTSPLIT



	A	B	C	D
1	First Last Name	First	Last	Name

The TEXTSPLIT function splits text into separate cells based on a specified delimiter. This is available in newer Excel versions like Excel 365 or Excel 2021.

Syntax: =TEXTSPLIT(text, col_delimiter, [row_delimiter])

Advance Numeric and Statistical Functions

SUMIF:

Formula bar: =SUMIF(B5:B15,"jim",D5:D15)

SUMIF function			
Name	State	Sales	
Jim	MN	\$100	
Sarah	CA	\$125	
Jane	GA	\$200	
Steve	CA	\$50	
Kelly	WA	\$125	
Walter	OR	\$75	
Brian	OR	\$100	
Jamie	CA	\$200	
Ayako	NV	\$250	
Jim	IA	\$100	
Joan	WA	\$150	

Criteria	Result	
Name = Jim	\$200	=SUMIF(B5:B15,"jim",D5:D15)
State = CA	\$375	=SUMIF(C5:C15,"ca",D5:D15)
Sales > \$100	\$1,050	=SUMIF(D5:D15,">100")

Finding the total sales of only rows that met the criteria

Adds values based on one condition.

=SUMIF(range, criteria, [sum_range])

SUMIFS:

Formula bar: =SUMIFS(F5:F15,C5:C15,"red",D5:D15,"TX")

SUMIFS function					
Date	Color	State	Qty	Total	
9-Jan	Red	TX	1	\$18.00	
23-Jan	Blue	CO	2	\$34.00	
3-Feb	Red	NM	2	\$36.00	
18-Feb	Blue	TX	1	\$17.00	
2-Mar	Blue	AZ	3	\$51.00	
15-Mar	Red	AZ	1	\$17.00	
25-Mar	Red	NV	2	\$34.00	
3-Apr	Blue	AZ	1	\$17.00	
11-Apr	Red	TX	2	\$34.00	
30-Apr	Blue	CO	1	\$17.00	
1-May	Red	TX	2	\$36.00	

Criteria	Result	
Red	\$175.00	
Red and TX	\$88.00	

Finding the total sales amount of

Adds values based on multiple conditions.

=SUMIFS(sum_range, criteria_range1, criteria1, ...)

AVERAGEIF:

H5									
	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									
12									
13									
14									
15									
16									

AVERAGEIF Function

Address	Price	Beds	Baths
3007 Arthur Ave	\$0	2	1
2479 North Rd	\$109,900	1	1
4318 D Street	\$112,000	2	1
4883 Hartland Ave	\$129,900	1	1
4150 Richland	\$149,900	2	1
2659 Crestview Ln	\$189,000	3	2
1448 Cheno Dr	\$229,900	4	2
1301 Robb Ct	\$355,000	3	2
4803 Hoffman Ave	\$385,000	4	2
897 Wiseman St	\$448,000	5	3
1780 Teak St	\$589,900	4	3

Criteria	Average
> \$0	\$269,850
> \$200k	\$401,560
2+ beds	\$273,189
3+ beds	\$366,133

Finding the Average of Price on each condition.

Averages values based on one condition.

=AVERAGEIF(range, criteria, [average_range])

AVERAGEIFS:

H5									
	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5									
6									
7									
8									
9									
10									
11									
12									
13									
14									
15									
16									

AVERAGEIFS Function

Address	Price	Beds	Baths
3007 Arthur Ave	\$0	2	1
2479 North Rd	\$109,900	1	1
4318 D Street	\$112,000	2	1
4883 Hartland Ave	\$129,900	1	1
4150 Richland	\$149,900	2	1
2659 Crestview Ln	\$189,000	3	2
1448 Cheno Dr	\$229,900	4	2
1301 Robb Ct	\$355,000	3	2
4803 Hoffman Ave	\$385,000	4	2
897 Wiseman St	\$448,000	5	3
1780 Teak St	\$589,900	4	3

Criteria	Average
> \$0	\$269,850
> \$0 and < \$500k	\$234,289
2+ beds and >1 baths	\$366,133

Finding the average of price on more

Averages values based on multiple conditions.

=AVERAGEIFS(average_range, criteria_range1, criteria1, ...)

COUNTIF:

G5

:

Count each row
where the condition
is TRUE.

Counts cells that meet one condition.

=COUNTIF(range, criteria)

COUNTIFS:

I6	:	X	✓	<i>f_x</i>	=COUNTIFS(C5:C16,"red",D5:D16,"TX")																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																			
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Count only rows with
more than one condition
should be true.

Counts cells that meet multiple conditions.

=COUNTIFS(criteria_range1, criteria1, ...)

MAXIFS:

G5 fx =MAXIFS(D5:D16,C5:C16,"F")

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											
17											

MAXIFS function

Name	Gender	Score
Hannah	F	93
Edward	M	79
Miranda	F	85
William	M	64
Joanna	F	81
Collin	M	83
Oscar	M	64
Arturo	M	76
Annie	F	72
Weston	M	64
Joshua	M	69
Cassidy	F	83

Gender	Highest
Female	93
Male	83

Find the Max and Min Height of each Gender.

Finds the max value based on conditions.

$\text{=MAXIFS}(\text{max_range}, \text{criteria_range1}, \text{criteria1}, \dots)$

MINIFS:

H7 fx =MINIFS(D3:D15,C3:C15,"F")

	A	B	C	D	E	F	G	H	I	J
1										
2										
3										
4										
5										
6										
7										
8										
9										
10										
11										
12										
13										
14										
15										

MINIFS(min_range,criteria_range,criteria,...)

Gender	Highest	Lowest
Female	93	72
Male	83	64

Finding the Min score of both genders.

Finds the min value based on conditions.

$\text{=MINIFS}(\text{min_range}, \text{criteria_range1}, \text{criteria1}, \dots)$

Advanced Lookup Functions

Lookup functions in Excel are used to search for specific data in a range or table and return corresponding values. These functions help retrieve data from large datasets based on matching criteria.

VLOOKUP:

The screenshot shows an Excel spreadsheet with the formula bar displaying `=VLOOKUP(G6,B6:E17,4,FALSE)`. The spreadsheet contains a table with the following data:

1	2	3	4
ID	First	Last	Email
610	Janet	Farley	janet.farley@acme.com
798	Steven	Batista	steven.batista@acme.com
841	Evelyn	Monet	evelyn.monet@acme.com
886	Marilyn	Bradley	marilyn.bradley@acme.com
622	Jonathan	Adder	jonathan.adder@acme.com
601	Adrian	Birt	adrian.birt@acme.com
869	Julie	Irons	julie.irons@acme.com
867	Erica	Tan	erica.tan@acme.com
785	Harold	Clayton	harold.clayton@acme.com

Below the table, a diagram illustrates the VLOOKUP function. It shows a 'Lookup Column' (ID) and a 'Result Column' (Email). An arrow points from the 'Lookup Column' to the 'Result Column'. To the right, a smaller table shows the lookup process:

ID	Email
869	julie.irons@acme.com

Arrows point from the 'Lookup Value' (869) to the 'Result' (julie.irons@acme.com).

Searches vertically in the first column of a range and returns a value from a specified column.

`=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`

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HLOOKUP:

D5 *x* *✓* *f_x* =HLOOKUP(C5,table,2,1) ←

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											

HLOOKUP function results

Name	Sales	Level	Bonus
Ainsley	171,900	Gold	5%
Bueller	205,350	Platinum	7%
Chung	151,200	Gold	5%
Crawford	119,850	Gold	5%
Joyce	89,450	Silver	3%
King	124,500	Gold	5%
Murray	131,100	Gold	5%
Richards	55,300	Silver	3%
Tanaka	201,500	Platinum	7%

Lookup values

Lookup table

Sales	50,000	100,000	175,000
Level	Silver	Gold	Platinum
Bonus	3%	5%	7%

table = H4:J6

Note lookup table is arranged horizontally

Searches horizontally in the first row of a range and returns a value from a specified row.

=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

INDEX:

G7 *x* *✓* *f_x* =INDEX(B5:E13,5,3) ←

	A	B	C	D	E	F	G	H
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								

INDEX function

	1	2	3	4
	Planet	Position	Diameter	Satellites
1	Mercury	1	4,879	0
2	Venus	2	12,104	0
3	Earth	3	12,756	1
4	Mars	4	6,792	2
5	Jupiter	5	142,984	67
6	Saturn	6	120,536	200
7	Uranus	7	51,118	27
8	Neptune	8	49,528	13
9	Pluto	9	2,306	5

142,984

Jupiter's diameter = row 5, column 3

Retrieves the value at a specific row and column in a range.

=INDEX(array, row_num, [column_num])

MATCH:

MATCH function
MATCH (lookup_value, lookup_array, match_type)

Fruit	
Apple	1
Pear	2
Grape	3
Lemon	4
Peach	5
Lime	6
Kiwi	7
Mango	8
Pineapple	9

Lookup	Result
Peach	5

Finds the position of a value in a range.

=MATCH(lookup_value, lookup_array, [match_type])

XLOOKUP:

XLOOKUP function
XLOOKUP (lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])

Planet	Satellites	Diameter (km)
Mercury	0	4,879
Venus	0	12,104
Earth	1	12,756
Mars	2	6,792
Jupiter	67	142,984
Saturn	200	120,536
Uranus	27	51,118
Neptune	13	49,528
Pluto	5	2,306

Planet	Diameter
Mars	6,792

lookup value

lookup array

return array

Searches for a value and returns a corresponding value from a different range, working both vertically and horizontally.

=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])

INDEX and MATCH Combine

H5 X ✓ fx =INDEX(item,MATCH(G5,id,0))

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											
17											

Left lookup with INDEX and MATCH

Item	Color	Price	ID
T-shirt	Red	\$18.00	1001
T-shirt	Blue	\$17.50	1002
T-shirt	Black	\$19.00	1003
Sandals	Gray	\$24.00	2001
Sandals	Black	\$25.00	2002
Hoodie	Black	\$33.00	3001
Hoodie	Gray	\$35.00	3001
Hoodie	White	\$32.00	3001
Hat	Black	\$28.00	4001
Hat	Orange	\$30.00	4002
Hat	White	\$25.00	4003

ID	Item	Color	Price
1003	T-shirt	Black	\$19.00
3001	Hoodie	Black	\$33.00
4002	Hat	Orange	\$30.00

item = B5:B15
color = C5:C15
price = D5:D15
id = E5:E15

INDEX and MATCH are often used together as an alternative to VLOOKUP or HLOOKUP for more flexibility.

4. LOOKUP

F6 X ✓ fx =LOOKUP(F4,B5:B9,C5:C9)

	A	B	C	D	E	F	G	H	I	J	K
1											
2											
3											
4											
5											
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											
17											

LOOKUP function

Level	Tier
1000	Bronze
2000	Silver
3000	Gold
4000	Platinum
5000	Diamond

Points	3125
Level	3000
Tier	Gold

// value matched in level
// corresponding value in tier

The LOOKUP function searches for a value in a range (row or column) and returns a corresponding value from another range.

Syntax:=LOOKUP(lookup_value, lookup_vector, [result_vector])

Error Handling in Excel

IFERROR

Excel formula bar: `=IFERROR(C5/D5,0)`

IFERROR function

Order #	Total	Qty	Avg price
1103	\$300.00	62	\$4.84
1104	\$14.00	0	\$0.00
1105	\$200.00	11	\$18.18
1106	\$120.00	7	\$17.14
1107	\$475.00	20	\$23.75
1108	\$360.00	30	\$12.00
1109	\$275.00	25	\$11.00
1110	\$400.00		\$0.00
1111	\$500.00	25	\$20.00
1112	\$600.00	50	\$12.00
1113	\$100.00	12	\$8.33

// avoid #DIV/0!

// avoid #DIV/0!

The Excel IFERROR function returns a custom result when a formula generates an error, and a standard result when no error is detected.

`=IFERROR(value,value_if_error)`

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Excel formula bar: `=IFNA(VLOOKUP(H8,$B$5:$C$16,2,FALSE),"")`

IFNA function	
Item	Price (\$)
Apple	0.50
Banana	0.30
Cherry	2.00
Orange	0.45
Pear	1.20
Grape	2.50
Honeydew	3.00
Plum	1.10
Pineapple	4.00
Kiwi	1.80
Lemon	0.60
Mango	2.80

Item	Price
Pear	1.20
Orange	0.45
Apple	0.50
Lime	#N/A

Without IFNA

Item	Price
Pear	1.20
Orange	0.45
Apple	0.50
Lime	

With IFNA

The Excel IFNA function is a simple way to handle #N/A errors specifically without catching other errors.

`=IFNA(value,value_if_na)`

That's Great

Data Study & Preparation Techniques

Data Study Techniques

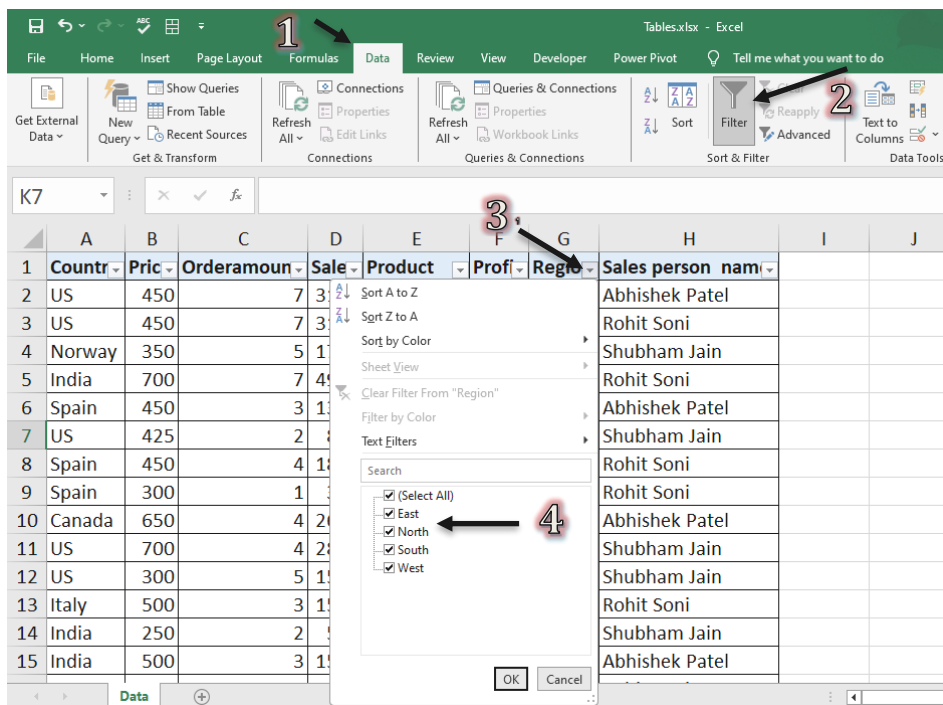
Data analysis techniques help users explore, visualize, and interpret data efficiently. Among these techniques, Conditional Formatting, Sorting, and Filtering are commonly used to study and analyse data.

Filters

Filtering helps you view specific subsets of data that meet certain criteria without altering or deleting any information. Filters enable you to focus on relevant data, removing distractions and helping you concentrate on specific patterns or relationships.

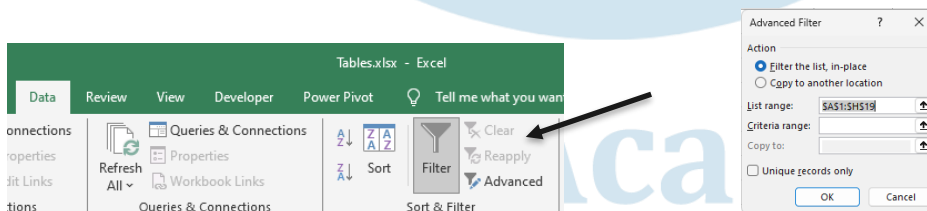
- Show rows based on specific conditions, such as all products sold in a particular region or transactions above a certain value.
- Apply multiple filters across different columns to narrow down the dataset further.

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1. Go to Data tab
2. Select Filter
3. Select the column you want to apply your filter criteria and click on drop down
4. Select the value you want to filter.

Advance Filter



When you want to filter the data on the bases of multiple criteria then use advance filter.

1. Filter the data where it is or copy it in new location

2. List range – Range where the data is
3. Criteria range – Use the same column and write your criteria on that column and select that range and click okay.

J	K	L	M	N	O	P	Q
Country	Price	Orderamount	Sales	Product	Profit	Region	Sales person name
					>=400	West	

	A	B	C	D	E	F	G	H
1	Country	Price	Orderamount	Sales	Product	Profit	Region	Sales person name
2	US	450	7	3150	Playstation	1200	West	Abhishek Patel
3	US	450	7	3150	Playstation	1200	West	Rohit Soni
7	US	425	2	850	Playstation	450	West	Shubham Jain
10	Canada	650	4	2600	Nintendo	400	West	Abhishek Patel

4. Now all the data have been sorted according to criteria.

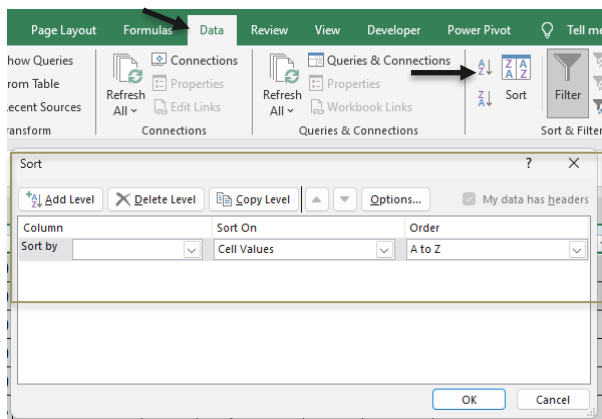
Sorting Data

Sorting is a simple yet essential technique for organizing data in Excel. It helps you arrange your data in a specific order, making it easier to analyse. There is basic sorting and advance sorting.

Basic Ascending or descending order: You can sort text alphabetically (A to Z or Z to A), numbers from smallest to largest (or vice versa), or dates in chronological order.

C	D	E	F	G
Orderamount	Sale	Product	Profit	Region
				West
				West
				North
				East

Advance Custom sorting: You can sort based on multiple columns, allowing for a more detailed organization of data (e.g., sorting sales by region and within each region by product).



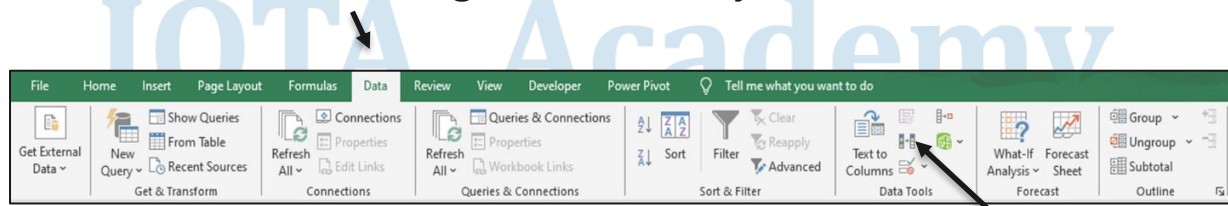
1. First select on Data -> Sort option
2. You can select a column then the value of cell that contain and then the sorting order like Ascending or Descending
3. You can even add a second level by adding Level option.

Data Preparation Techniques

Remove Duplicates

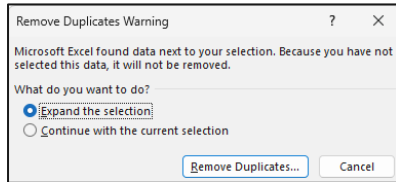
Remove Duplicates is an Excel feature that deletes duplicate rows from a dataset, keeping only unique entries.

Use: It helps in cleaning up data by removing redundant information and ensuring that each entry is distinct.



1. To use it first select the range which have duplicated values and then go to

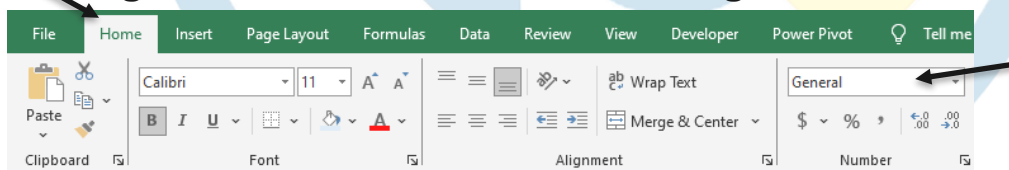
Data -> Remove duplicates.



2. If you have multiple columns of data, it will ask if you remove from single column or current selection or from whole data (Expand selection).

Checking & Applying Correct Data Type

Applying the correct data type ensures accurate calculations, proper sorting and filtering, and correct interpretation of data. It prevents errors like miscalculations or incorrect formula results, leading to reliable data analysis and insights, which are crucial for making informed decisions and avoiding mistakes.

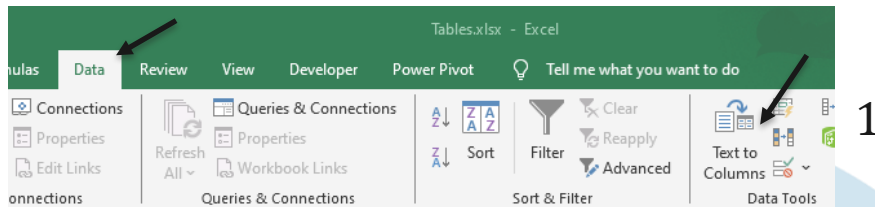


Select the column and go to Home and select the correct data type for the data in a column.

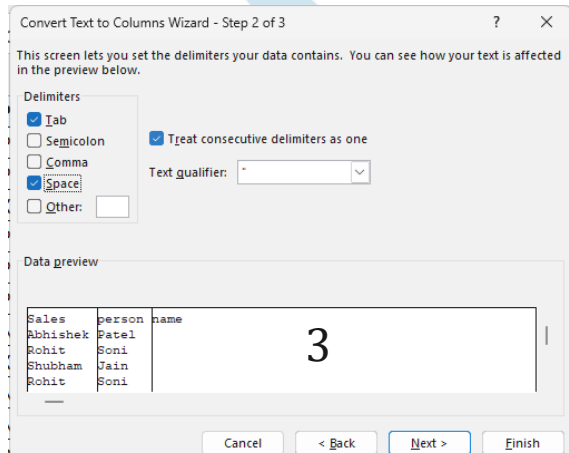
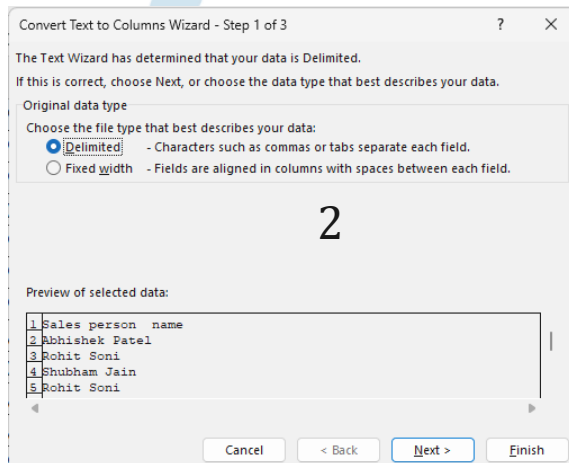
Text To Column

The Text to Columns feature in Excel splits data in a single column into multiple columns based on a specified delimiter (such as commas, spaces, or custom characters) or a fixed width. This is

useful for separating combined data, like splitting full names into first and last names or parsing addresses into different parts.



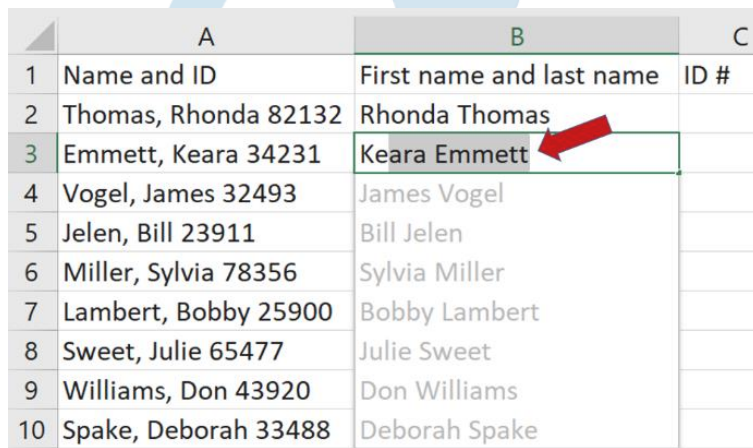
1. Select the column which you want to split.
2. Go to Data tab -> Text to column



3. Select the delimiter and finish. Column will get split.

Flash Fill

Flash Fill is a tool that automatically fills in data based on a **pattern it recognizes** from user input. It can be used to format, extract, or combine data across columns without needing formulas. ***For example***, if you're splitting names or combining data, Flash Fill quickly identifies the pattern and completes the task for you.

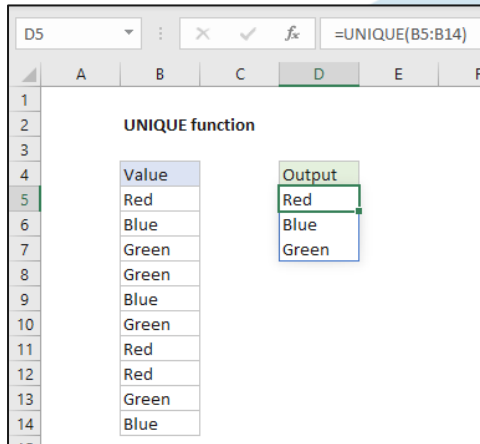


	A	B	C
1	Name and ID	First name and last name	ID #
2	Thomas, Rhonda 82132	Rhonda Thomas	
3	Emmett, Keara 34231	Keara Emmett	
4	Vogel, James 32493	James Vogel	
5	Jelen, Bill 23911	Bill Jelen	
6	Miller, Sylvia 78356	Sylvia Miller	
7	Lambert, Bobby 25900	Bobby Lambert	
8	Sweet, Julie 65477	Julie Sweet	
9	Williams, Don 43920	Don Williams	
10	Spake, Deborah 33488	Deborah Spake	

- After typing the pattern for a few rows, Excel might automatically detect the pattern and fill in the rest of the column.
- If it doesn't, manually activate Flash Fill by selecting the next cell where the pattern should continue, and press **Ctrl + E** or go to the **Data** tab and click **Flash Fill**.

UNIQUE Function

The UNIQUE function in Excel returns a list of unique values from a range or array. It eliminates duplicates, allowing you to quickly identify distinct entries in a dataset. **(Work on Excel 2019 or later)**

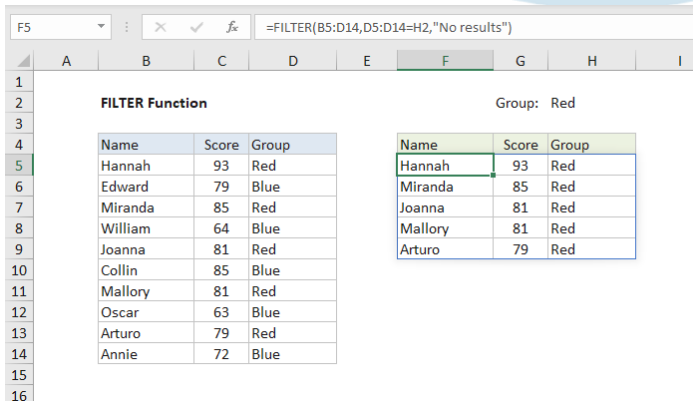


Value	Output
Red	Red
Blue	Blue
Green	Green
Green	
Blue	
Green	
Red	
Red	
Green	
Blue	

Syntax: =UNIQUE(array, [by_col], [exactly_once])

FILTER Function

The FILTER function in Excel extracts data that meets specific criteria. It allows you to filter a dataset dynamically based on conditions, making it easier to isolate and analyze relevant information. **(Work on Excel 2019 or later)**



Name	Score	Group
Hannah	93	Red
Edward	79	Blue
Miranda	85	Red
William	64	Blue
Joanna	81	Red
Collin	85	Blue
Mallory	81	Red
Oscar	63	Blue
Arturo	79	Red
Annie	72	Blue

Syntax: =FILTER(array, include, [if_empty])

That's Great



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What is Data Visualization

Data visualization is the graphical representation of data using charts, graphs, and other visual tools to make information easier to understand. It helps in identifying patterns, trends, and outliers, making complex data more accessible and insightful for decision-makers. Common forms include bar charts, line graphs, pie charts, and heat maps. Effective data visualization simplifies data interpretation, allowing for quicker analysis and more informed, data-driven decisions.

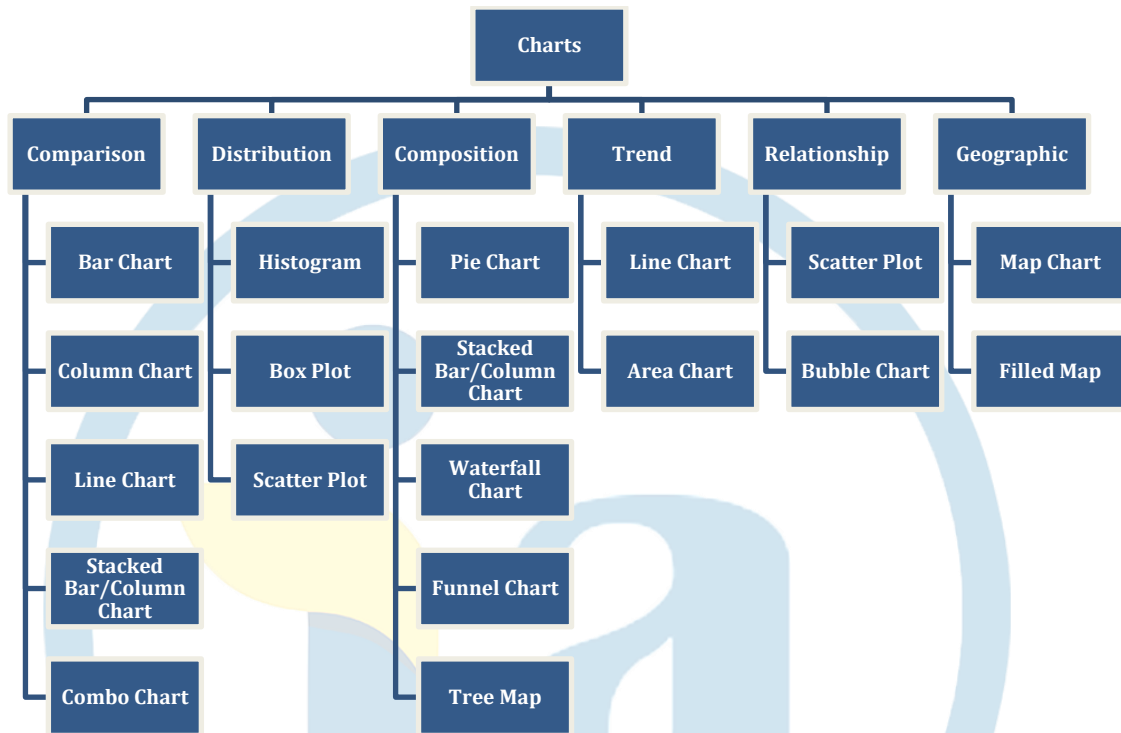
Importance of Data Visualization

Data visualization is crucial because

- It transforms complex data into intuitive, visual formats, making it easier to understand and interpret.
- It helps identify patterns, trends, and outliers quickly, enabling faster decision-making.
- Visuals enhance storytelling, making data insights more accessible to both technical and non-technical audiences.
- It improves communication, reveals hidden relationships in data, and aids in better strategic planning, ultimately leading to more informed, data-driven decisions.

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Types of visualizations

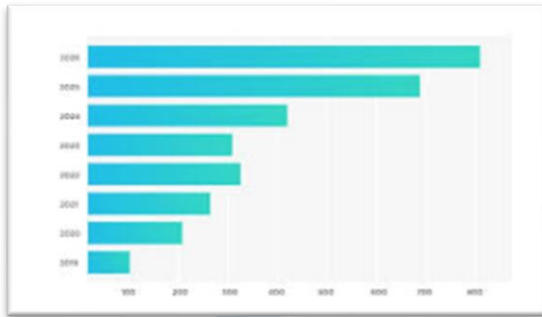


Comparison Charts

Used to compare different data sets, categories, or values. Bar, column, and line charts fall into this group, showing side-by-side comparisons of variables. They are ideal for spotting trends, similarities, or differences between data points, making them essential for performance evaluation and analysis.

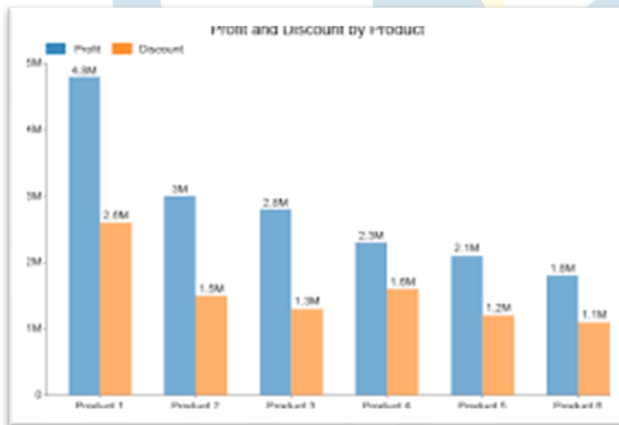
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- Bar chart



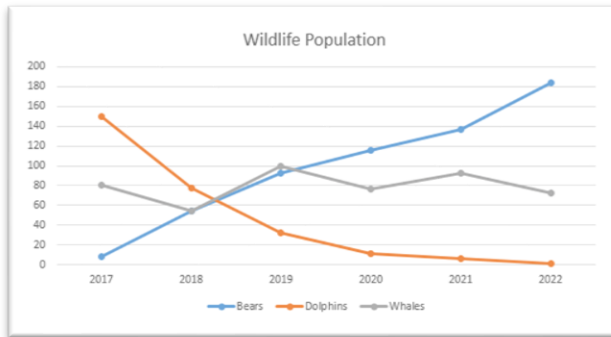
A Bar Chart is a type of chart that represents categorical data with rectangular bars, where the length of each bar is proportional to the value it represents.

- Column chart



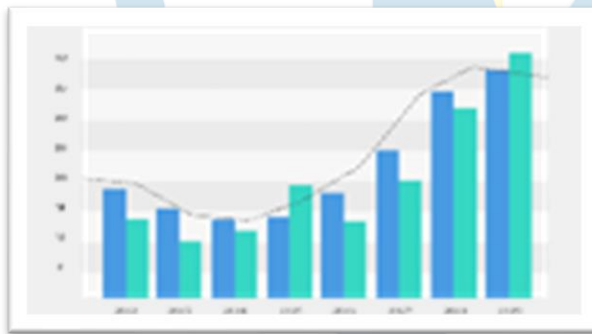
A Column Chart is similar to a bar chart but displays data with vertical bars, where the height of each bar represents the value of the data point.

- Line chart



A Line Chart uses points connected by straight lines to represent data trends over time. Each point on the chart corresponds to a specific value, typically plotted on the vertical axis, while the horizontal axis represents time or another continuous variable.

- Combo chart



A Combo Chart combines bar/column chart and a line chart, in one visualization. This allows the comparison of different types of data or metrics within the same chart.

- Table

Year	Quarter	Month	Day	Sum of High
2016	Qtr 3	July	11	97.65
2016	Qtr 3	July	12	97.70
2016	Qtr 3	July	13	97.67
2016	Qtr 3	July	14	98.99
2016	Qtr 3	July	15	99.30
2016	Qtr 3	July	16	100.13
2016	Qtr 3	July	19	100.00
2016	Qtr 3	July	20	100.46
2016	Qtr 3	July	21	101.00
2016	Qtr 3	July	22	99.30
2016	Qtr 3	July	25	98.84
2016	Qtr 3	July	26	97.97
2016	Qtr 3	July	27	104.35
2016	Qtr 3	July	28	104.45
2016	Qtr 3	July	29	104.55
2016	Qtr 3	August	1	106.15
2016	Qtr 3	August	2	106.07
2016	Qtr 3	August	3	105.84
2016	Qtr 3	August	4	106.00
2016	Qtr 3	August	5	107.65
2016	Qtr 3	August	8	108.37
2016	Qtr 3	August	9	108.94
2016	Qtr 3	August	10	108.90
2016	Qtr 3	August	11	108.93
2016	Qtr 3	August	12	108.44
Total				51,476.68

Table Chart is a visualization that displays data in a grid format with rows and columns. It allows for detailed tabular presentation of data, showing values and metrics with headers for each column.

- Matrix

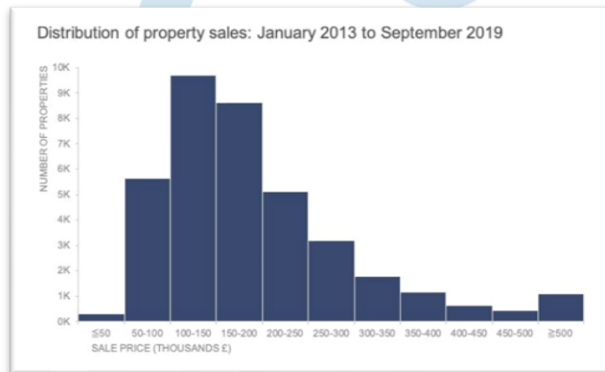
Region	Sales Stage	Opportunity Count	Revenue	Sales	Opportunity Count	Revenue	West	Opportunity Count	Revenue	Total	Opportunity Count	Revenue
Lead												
Small		26	\$11,980,676	16	\$47,418,106	14	\$11,891,016	75	\$62,226,488			
Medium		25	\$76,340,147	30	\$71,131,216	18	\$72,871,037	73	\$248,846,140			
Large		40	\$323,570,192	23	\$223,348,275	10	\$149,426,712	91	\$1,022,685,180			
Total		91	\$445,490,915	69	\$441,957,497	42	\$234,195,765	239	\$1,534,557,808			
Qualified												
Small		10	\$71,530,085	10	\$27,905,216	5	\$2,489,889	25	\$101,925,110			
Medium		12	\$40,030,528	16	\$71,617,018	8	\$53,045,360	36	\$165,492,500			
Large		7	\$21,144,920	12	\$101,189,888	2	\$12,127,426	21	\$134,452,200			
Total		29	\$132,705,533	38	\$170,712,122	15	\$67,662,675	82	\$302,869,810			
Solution												
Small		13	\$12,275,131	8	\$10,283,036	7	\$2,155,181	28	\$24,713,348			
Medium		9	\$20,940,340	11	\$34,279,217	4	\$35,163,417	24	\$70,418,000			
Large		7	\$40,362,132	9	\$39,313,963	4	\$29,862,847	20	\$109,539,000			
Total		29	\$153,577,503	28	\$174,876,216	15	\$67,181,445	72	\$204,670,348			
Proposal												
Small		8	\$11,096,116	7	\$4,711,862	2	\$2,165,031	17	\$18,973,009			
Medium		4	\$15,283,151	6	\$25,687,381	5	\$21,458,037	15	\$62,447,670			
Large		2	\$16,344,322	4	\$20,164,481	2	\$12,483,443	8	\$59,192,440			
Total		14	\$42,723,589	17	\$50,563,724	9	\$36,106,511	39	\$138,613,119			
Finalize												
Small		1	\$1,786,107	1	\$1,691,185	2	\$7,325,517	4	\$16,802,925			
Medium		2	\$2,979,200	4	\$2,957,681	2	\$15,497,640	8	\$16,750,480			
Large		7	\$12,180,900	5	\$26,496,420	4	\$21,176,185	16	\$75,174,800			
Total		10	\$16,946,207	10	\$31,145,286	8	\$44,009,342	28	\$108,728,205			

Matrix Chart is a flexible table-like visualization that displays data in a grid with rows and columns, similar to a pivot table. It allows for dynamic aggregation and grouping of data, with the ability to nest multiple levels of rows and columns.

Distribution Charts

Visualize the spread of data points to reveal the range, outliers, and frequency of occurrences. Examples include histograms and scatter plots. These charts are useful for understanding the distribution patterns in datasets, showing concentrations of data, and identifying any anomalies or unusual patterns.

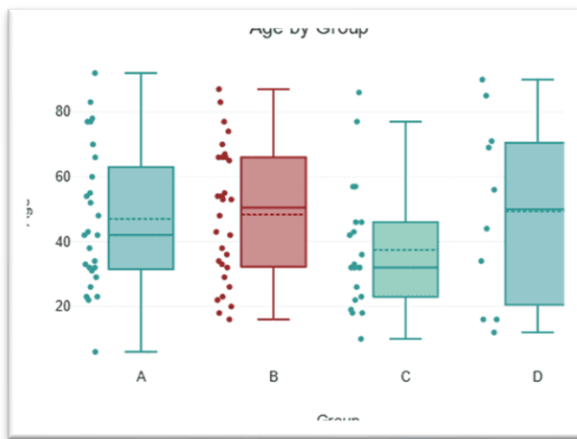
- Histogram



A Histogram is a type of chart that displays the distribution of numerical data by grouping data points into ranges, or "bins." Each bar represents the frequency of data points within a specific range.

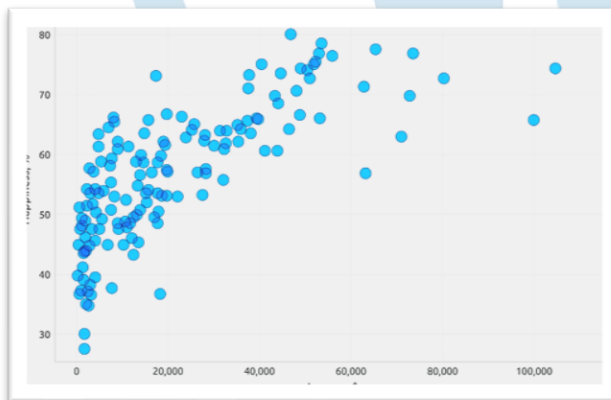
- Box plot

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A Box Plot (or Box-and-Whisker Plot) is a chart that visualizes the distribution, central tendency, and variability of a dataset. It shows five key statistics: minimum, first quartile (Q1), median, third quartile (Q3), and maximum. The "box" represents the interquartile range (Q1 to Q3), and the "whiskers" extend to the minimum and maximum values.

- Scatter Plot Chart

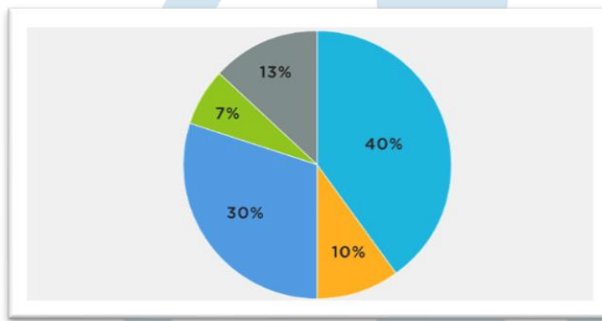


A Scatter Plot is a chart that uses dots to represent the values of two different variables. Each dot's position on the horizontal and vertical axes indicates the values for those two variables, allowing you to visualize the relationship between them.

Composition Charts

Display how a whole is divided into parts, showing contributions of individual components. Pie charts, stacked bar/column charts, and tree maps are examples. They help understand proportions and relative percentages, making it easy to assess which components are more or less significant in the dataset.

- Pie chart



A Pie Chart is a circular chart divided into slices, where each slice represents a proportion of the whole. The size of each slice is proportional to the value or percentage it represents within the dataset.

- Tree Map



A Tree Map is a chart that displays hierarchical data using nested rectangles. Each rectangle represents a category, and its size is

proportional to the value it holds. Subcategories can be shown as smaller rectangles within the larger ones, visually representing the hierarchical structure.

- Waterfall chart

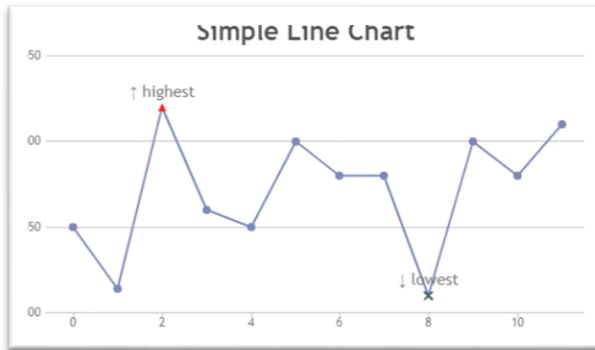


A Waterfall Chart is a type of data visualization that shows incremental changes between initial and final values, often used to display how a starting value is influenced by various factors over a period. The chart consists of bars representing positive and negative changes, which are connected to illustrate the cumulative impact on the total value.

Trend Charts

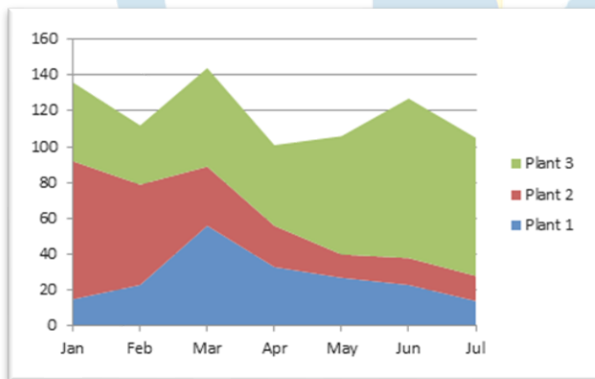
Depict data over time to show upward, downward, or stable patterns. Line and area charts are typical examples. Trend charts are used to track changes, forecast future developments, or analyze periodic behaviors, helping users spot long-term tendencies and fluctuations in a dataset.

- Line chart



A Line Chart uses points connected by straight lines to represent data trends over time. Each point on the chart corresponds to a specific value, typically plotted on the vertical axis, while the horizontal axis represents time or another continuous variable.

- Area Chart

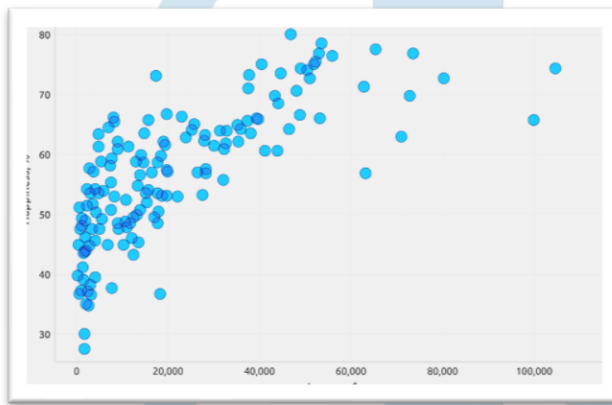


An Area Chart is similar to a line chart but with the area below the line filled with color or shading. It shows trends over time by plotting data points and filling the space between the line and the axis, highlighting the volume of data.

Relationship Charts

Display connections between two or more variables, helping identify correlations or interactions. Scatter plots and bubble charts are common examples. These charts are ideal for analyzing how one factor affects another, making it easier to spot associations, dependencies, or anomalies in multi-variable data.

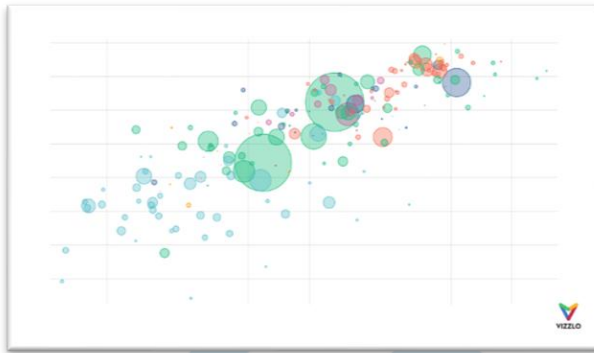
- Scatter Plot Chart



A Scatter Plot is a chart that uses dots to represent the values of two different variables. Each dot's position on the horizontal and vertical axes indicates the values for those two variables, allowing you to visualize the relationship between them.

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- Bubble Chart



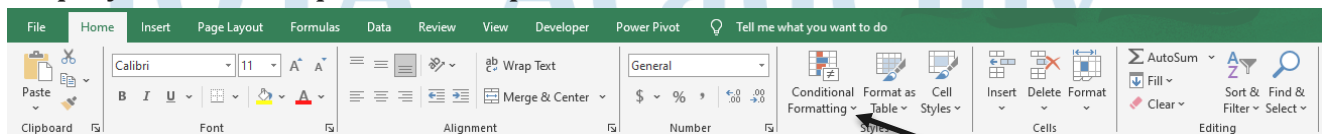
Bubble Chart is a type of scatter plot where each data point is represented by a bubble. In addition to showing the relationship between two variables (x and y axes), the size of each bubble represents a third variable, adding an extra dimension to the visualization.

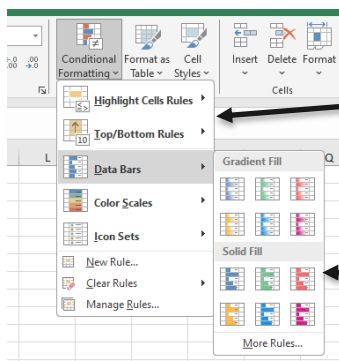
Conditional Formatting:

Conditional formatting is a powerful feature that allows you to apply specific formatting to cells based on their values. It helps in visually identifying patterns, trends, and outliers in a dataset. For example,

- Cells with values above or below a certain threshold.
- Duplicate values or unique entries.
- Cells based on data bars, color scales, or icon sets to represent variations.

With the help of conditional formatting, you can quickly spot critical information, such as top-performing sales regions, low-performing employees, or unexpected data points that need attention.





Excel Conditional Formatting

Month	Price
Jan	\$120.00
Feb	\$60.00
Mar	\$80.00
Apr	\$102.49
May	\$70.00
Jun	\$99.12
Jul	\$54.00
Aug	\$105.79
Sep	\$45.00
Oct	\$70.00
Nov	\$97.52
Dec	\$105.94

Applying Conditional formatting on dataset:

- Select the range that you want to format
- Click on the Conditional Formatting icon in the ribbon, from the Home menu
- Select Color Scales from the drop-down menu

Thank you

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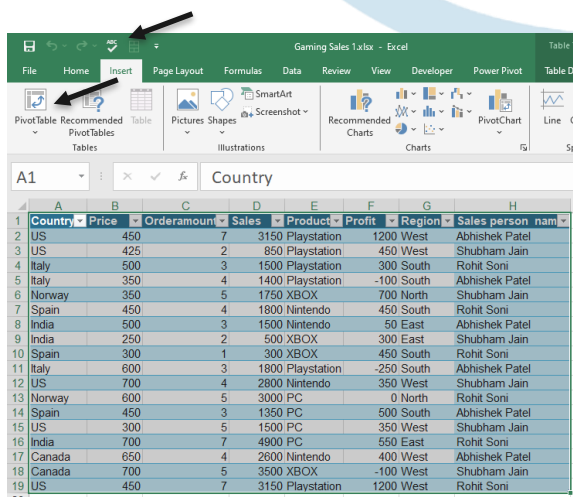
What is Pivot Table?

A Pivot Table is a tool that helps you quickly summarize large amounts of data. It lets you group and rearrange information, like sales or expenses, to get totals, averages, or counts without using complex formulas. You can easily drag and drop columns or rows to see the data in different ways, making it simple to analyze and spot trends or patterns.

What are the advantages of Pivot Table?

- **Quick Data Summarization:** Easily summarize large datasets by totals, averages, counts, etc.
- **Flexibility:** Drag and drop fields to rearrange data and view it from different perspectives.
- **Time-Saving:** Fast analysis without complex formulas.
- **Interactive Reports:** Update automatically when data changes.
- **Data Filtering:** Use filters to focus on specific parts of your data.

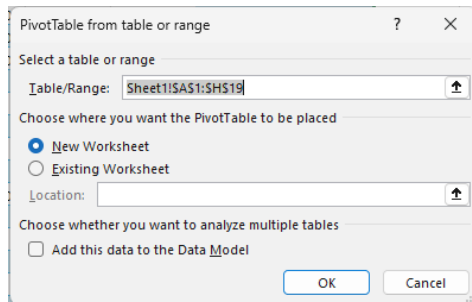
How to create a Pivot Table?



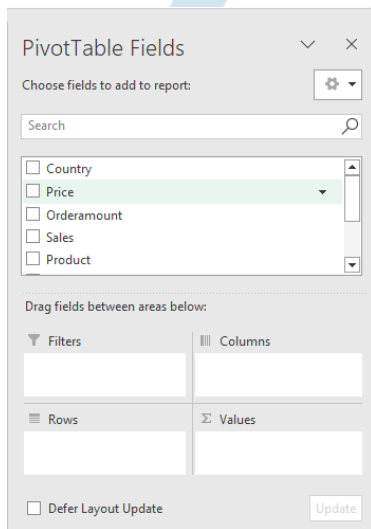
The screenshot shows the Microsoft Excel interface with the 'Insert' tab selected. The 'PivotTable' button is highlighted in the 'Tables' group. Below the ribbon, a PivotTable is displayed with 'Country' as the filter. The data is summarized by 'Region' and 'Sales person name'.

Country	Price	Order amount	Sales	Product	Profit	Region	Sales person name
US	450	7	3150	Playstation	1200	West	Abhishek Patel
US	425	2	850	Playstation	450	West	Shubham Jain
Italy	500	3	1500	Playstation	300	South	Rohit Soni
Italy	350	4	1400	Playstation	-100	South	Abhishek Patel
Norway	350	5	1750	XBOX	700	North	Shubham Jain
Spain	450	4	1800	Nintendo	450	South	Rohit Soni
India	500	3	1500	Nintendo	50	East	Abhishek Patel
India	250	2	500	XBOX	300	East	Shubham Jain
Spain	300	1	300	XBOX	450	South	Rohit Soni
Italy	600	3	1800	Playstation	-250	South	Abhishek Patel
US	700	4	2800	Nintendo	350	West	Shubham Jain
Norway	600	5	3000	PC	0	North	Rohit Soni
Spain	450	3	1350	PC	500	South	Abhishek Patel
US	300	5	1500	PC	350	West	Shubham Jain
India	700	7	4900	PC	550	East	Rohit Soni
Canada	650	4	2600	Nintendo	400	West	Abhishek Patel
Canada	700	5	3500	XBOX	-100	West	Shubham Jain
US	450	7	3150	Playstation	1200	West	Rohit Soni

1. First select the data range and go to insert -> Pivot Table

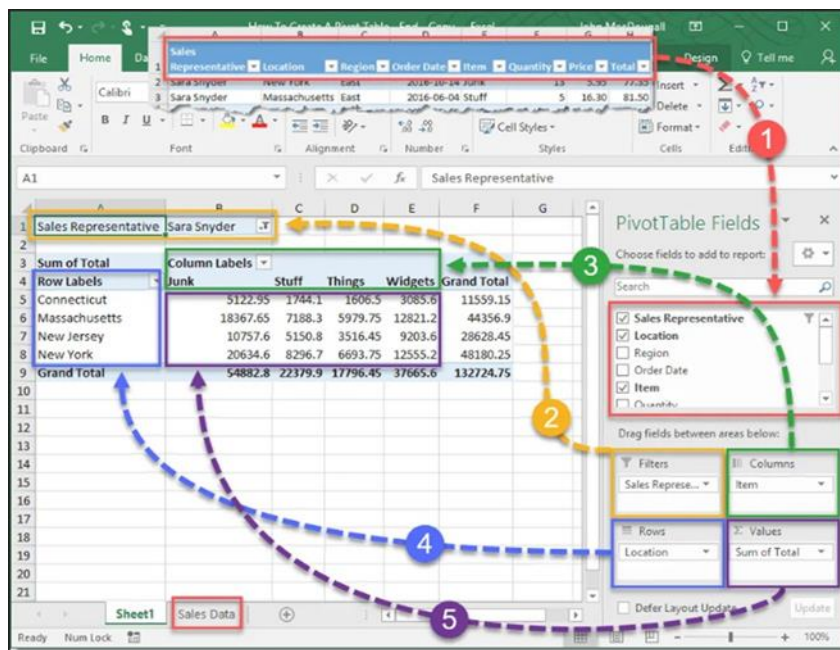


2. Confirm the range and select new sheet to make a pivot table in new sheet.



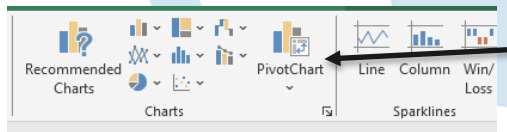
3. There are four fields to create a pivot table those are.

- **Rows:** Data placed here forms the row labels. These are the categories or groups (e.g., Product names, Regions).
- **Columns:** Data in this field forms the column labels (e.g., Months, Departments).
- **Values:** This is where the numerical data goes (e.g., Sales figures, Quantity sold). Excel will calculate sums, averages, or counts for the data placed here.
- **Filters:** Allows you to filter the entire Pivot Table based on specific criteria (e.g., filtering by Year or Salesperson).

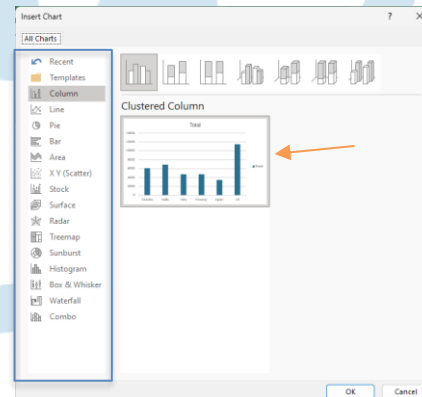


4. You can now drag and drop the fields and create a pivot Table.

Create a Pivot Chart



F	G	H	I
	Row Labels	Sum of Sales	
	Canada	6100	
	India	6900	
	Italy	4700	
	Norway	4750	
	Spain	3450	
	US	11450	
	Grand Total	37350	



- **Select the Pivot Table:** Click anywhere inside your Pivot Table.
- **Insert Pivot Chart:** Go to the Insert tab again, and select PivotChart from the Charts group.
- **Choose Chart Type:** Select your preferred chart type (e.g., Bar Chart, Pie Chart, Line Chart, etc.), then click OK.

Copying a Pivot Table

Step 1: Click anywhere inside the Pivot Table.

Step 2: Press Ctrl + C to copy the Pivot Table.

Step 3: Select the location where you want to paste it (new sheet or location in the same sheet).

Step 4: Press Ctrl + V to paste the Pivot Table.

(The copied Pivot Table remains connected to the same data source, but you can modify it independently from the original.)

Moving a Pivot Table

Step 1: Click anywhere inside the Pivot Table.

Step 2: On the PivotTable Tools ribbon, go to the Analyze tab (or Options tab in some versions).

Step 3: Click Move PivotTable in the Actions group.

Step 4: Choose a new location in the existing worksheet or select a new worksheet, then click OK.

Deleting a Pivot Table

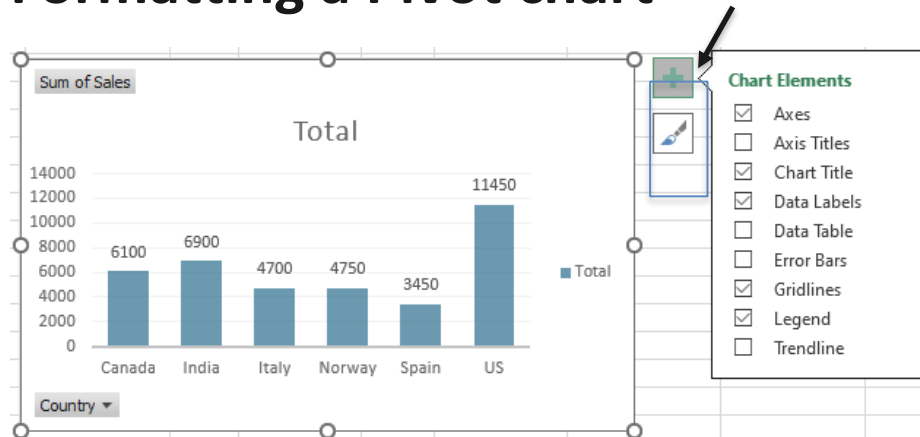
Step 1: Click anywhere inside the Pivot Table.

Step 2: Press Ctrl + A to select the entire Pivot Table.

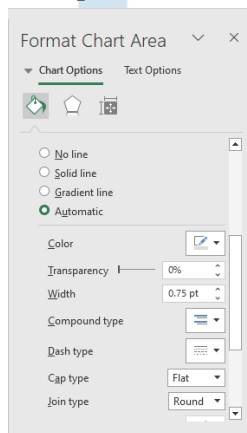
Step 3: Press Delete on your keyboard.

(Deleting a Pivot Table will not remove the underlying data source. Only the summary created by the Pivot Table is removed.)

Formatting a Pivot chart



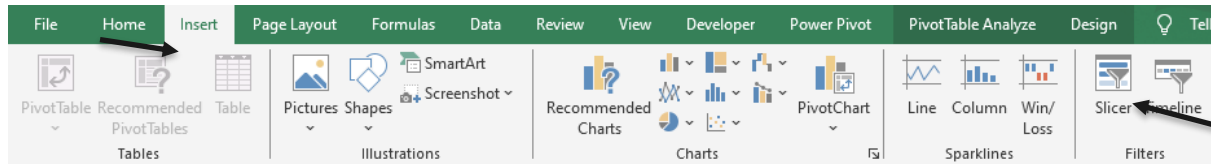
- You can add or remove chart elements and change the theme color of chart using option on top right of the chart.
- To format a chart on a detailed manner double click on a chart to open a new formatting pane on the right side of the screen.



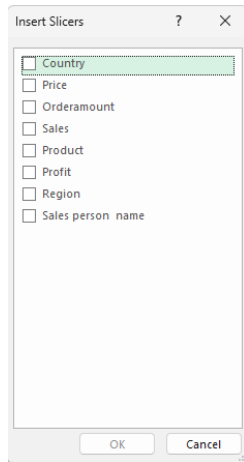
What is a Slicer?

Slicer is a interactive visual filters used to filter data within Excel tables and PivotTables. It provides a user-friendly method to concentrate on particular sections of your data, simplifying the analysis process.

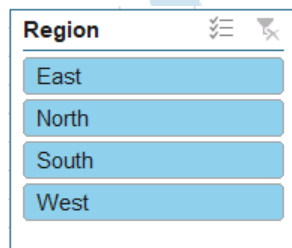
How to insert a slicer?



- First select the pivot table and then go to **Insert -> Slicer**.



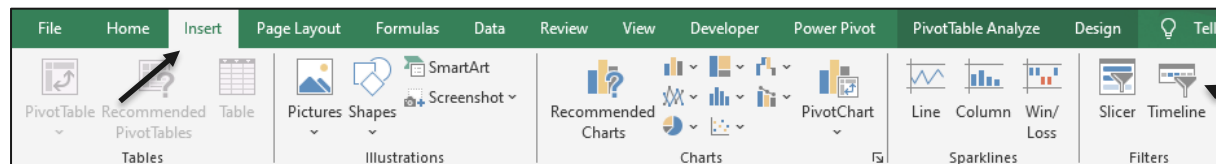
→ Select the column on the bases of which you want to slice your chart and tables.

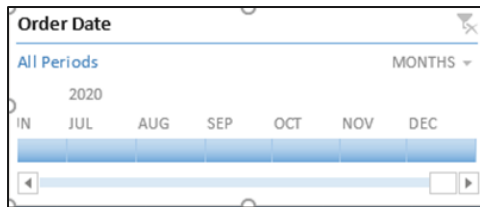


→ You will get a Slicer, by which you can just click on any value and the data will get slice/filtered.

Timeline

Timelines work much like slicers but will help to filter only based on the date/time field. It also allows you to zoom in and out on a particular period with the help of controllers.

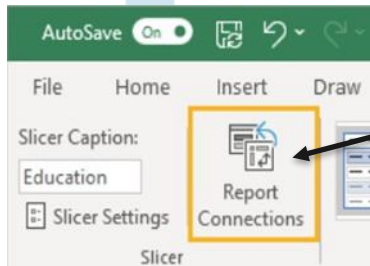




- You can select a time range to filter the data on the bases of dates.

Report Connection

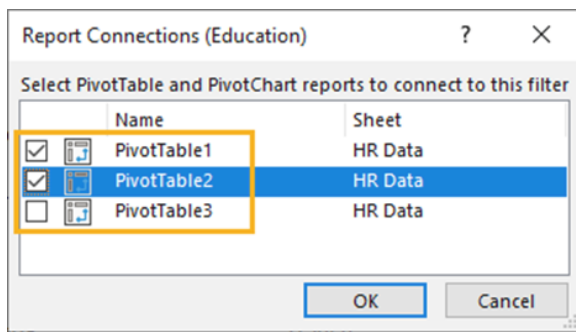
Report Connection in Slicers allows a single slicer to control multiple Pivot Tables or Pivot Charts that share the same data source. This feature helps you synchronize the filtering across different reports or dashboards to maintain consistency and avoid manually updating each Pivot Table or Pivot Chart separately.



You can open a slicer or timeline's Report Connection menu from two places.

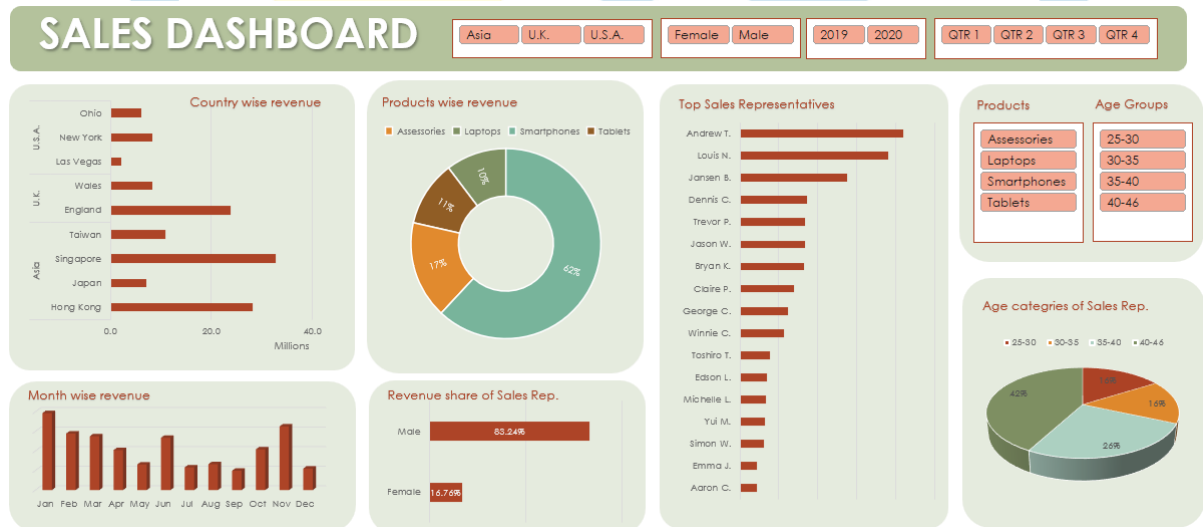
- Go to the Slicer or Timeline tab and select Report Connections.
- Right click on the slicer or timeline and go to Report Connections.

In the Report Connections menu, you can then select all the pivot tables in the workbook which you want to connect to the slicer.



Dashboard

After creating all the necessary chart and slicers you can copy past them in a new sheet and arrange them will make a dashboard.



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Thank you

What is VBA

VBA stands for Visual Basic for Applications. It's a programming language developed by Microsoft, primarily used for automating tasks and building custom functions in Microsoft Office applications like Excel, Word, and Access.

VBA vs VB

Visual Basic (VB) is a standalone programming language used to develop full-fledged Windows applications with the ability to create **.exe** and **.dll** files, whereas **Visual Basic for Applications (VBA)** is an embedded scripting language within Microsoft Office applications, primarily used for automating tasks and macros in Excel, Word, and Access. VB runs in Visual Studio and has unrestricted access to system resources, while VBA runs inside the Office environment with limited access for security reasons. VB focuses on application development, while VBA is designed for enhancing and automating Office workflows.

What are Macros

A macro is a set of recorded or written instructions that automate repetitive tasks in applications like Microsoft Excel, Word, and Access. In VBA (Visual Basic for Applications), macros are scripts that execute predefined actions, such as formatting cells, generating reports, or processing data, without manual input. Users can create macros by recording actions or writing VBA code in the VBA Editor.

VBA key components

Editor: Where you write and edit your code. It's part of the Office application, often found under the Developer tab.

Modules: These are containers for your VBA code. You can have multiple modules within a project.

Procedures: These are the blocks of code that perform specific tasks. They can be Sub procedures (which do something) or Function procedures (which return a value).

Variables: Used to store data that your code can manipulate.

Controls: These are elements like buttons, text boxes, and labels that you can add to forms.

What are Procedures

Procedures in VBA are blocks of code that perform specific tasks. They are the building blocks of your VBA programs. There are two main types of procedures:

Sub Procedures (Subroutines)

Sub procedure is a block of code that performs actions but doesn't return a value. It starts with the **Sub keyword** and ends with **End Sub**. Sub procedures can be called to execute tasks within your program.

```
Sub SayHello()  
    MsgBox "Hello, World!"  
End Sub
```

Function Procedures

Function procedure in VBA is a block of **code that performs a set of tasks and returns a value**. It starts with the Function keyword and ends with End Function.

```
Function AddNumbers(x As Integer, y As Integer) As Integer
    AddNumbers = x + y
End Function
```

What is an Object

An object is a "thing" just like real world things that you can manipulate through code. Objects represent elements in your application, such as worksheets, cells, ranges, and charts in Excel all of these are Objects in Excel. Each object has its own properties and methods.

Properties

Properties are attributes of an object. **They define characteristics or qualities of the object**. Think of them as the adjectives that describe something like font size, colour in a cell etc.

Methods

Methods are actions that can be performed on an object. They're like the verbs that tell what the object can do.

How to access any object in Excel VBA

To access any object, you have to use a dot (.) It is used to access the properties, methods, and events of an object. Think of it as a bridge that connects you to the object's features.

Example :- If you want to change the value of a cell in Excel, you'd navigate from the workbook to the worksheet to the range:

ThisWorkbook.Sheets("Sheet1").Range("A1").Value = "Hello, world!"

- **ThisWorkbook** refers to the workbook where the VBA code is being executed, The file you are working on.
- **Sheets("Sheet1")** accesses the "Sheet1" worksheet.
- **Range("A1")** specifies cell A1 in that worksheet.
- **Value** accesses the value property of the range object and sets it to "Hello, world!".

By chaining these components with dots, you can drill down to the specific element you want to manipulate. It's like following a path from the general to the specific.

Range Object

The Range object in VBA represents a cell or a range of cells in an Excel worksheet. It's a fundamental part of VBA for Excel, allowing you to manipulate cells' values, properties, and methods.

Range("A1:A10") = 34 -> this will insert 34 in all 10 cells

Methods to format Range

Here are some of the most commonly used methods for formatting a range in VBA:

Font:

Bold: Makes the font bold.

Range("A1:A10").Font.Bold = True

Italic: Makes the font italic.

Range("A1:A10").Font.Italic = True

Size: Changes the font size.

Range("A1:A10").Font.Size = 14

Interior:

Color: Changes the background color.

Range("A1:A10").Interior.Color = RGB(200, 200, 255)

Pattern: Applies a pattern.

Range("A1:A10").Interior.Pattern = xlPatternChecker

Borders:

LineStyle: Sets the border style.

Range("A1:A10").Borders.LineStyle = xlContinuous

Color: Sets the border color.

Range("A1:A10").Borders.Color = RGB(0, 0, 0)

Format

NumberFormat: Changes the number format of the cells.

Range("A1:A10").NumberFormat = "0.00"

Alignment

HorizontalAlignment: Aligns the text horizontally.

Range("A1:A10").HorizontalAlignment = xlCenter

VerticalAlignment: Aligns the text vertically.

Range("A1:A10").VerticalAlignment = xlCenter

WrapText: Wraps the text within the cell.

Range("A1:A10").WrapText = True

Merge: Merges the cells in the range.

Range("A1:A10").Merge

IF, Then, Elseif

The If...Then...Elseif statement in VBA allows for multiple conditions to be tested in sequence, executing different blocks of code depending on which condition is true. It's useful for handling complex decision-making processes in your code.

```
Sub CheckGrade()  
    Dim score As Integer  
    score = InputBox("Enter the score:")  
  
    If score ≥ 90 Then  
        MsgBox "Grade: A"  
    ElseIf score ≥ 80 Then  
        MsgBox "Grade: B"  
    ElseIf score ≥ 70 Then  
        MsgBox "Grade: C"  
    ElseIf score ≥ 60 Then  
        MsgBox "Grade: D"  
    Else  
        MsgBox "Grade: F"  
    End If  
End Sub
```

Explanation:

- If score >= 90 Then: Checks if the score is 90 or higher. If true, displays "Grade: A".
- ElseIf score >= 80 Then: If the first condition is false, checks if the score is 80 or higher. If true, displays "Grade: B".

- Elself score >= 70 Then: If the previous conditions are false, checks if the score is 70 or higher. If true, displays "Grade: C".
- Elself score >= 60 Then: If the previous conditions are false, checks if the score is 60 or higher. If true, displays "Grade: D".
- Else: If none of the conditions are true, displays "Grade: F".

With Block

The With statement in VBA is used to execute a series of statements on a single object without repeating the object's name.

```
Sub FormatCells()  
    With Range("A1:A10")  
        .Font.Bold = True  
        .Font.Size = 12  
        .Interior.Color = RGB(200, 200, 255)  
    End With  
End Sub
```

Instead of repeating the range object for font, size, color we use a with statement and under that use further objects at the end we end the with claus.

For Next Loop

The For...Next loop is a control structure used to repeat a block of code a specific number of times.


```

Sub SumNumbers()
    Dim num As Integer
    Dim i As Integer
    num = 0
    For i = 1 To 10
        num = num + i
    Next i
    MsgBox "The total sum is: " & num
End Sub

```

- Declares the num and i variables as integers.
- Initializes num to 0.
- Uses a For...Next loop to iterate from 1 to 10.
- Adds the value of i to num in each iteration.
- Displays the total sum in a message box.

You can also use an additional argument in for loop STEP to step count the number it is used just after the for loop in a same line.

Nested for loop

A nested For loop is simply a For loop inside another For loop. It's useful for tasks that require multiple levels of iteration, like populating a 2D grid or matrix.

```

Sub randuom()
    For i = 1 To 5
        For j = 1 To 10
            Cells(j, i) = Round(Rnd * 100, 0)
        Next j
    Next i
End Sub

```



	A	B	C	D	E
1	34	29	13	76	5
2	18	28	81	11	19
3	49	1	87	98	51
4	47	66	82	55	79
5	52	36	32	51	93
6	39	75	68	3	53
7	91	78	16	98	82
8	40	97	46	57	6
9	46	80	2	63	73
10	62	57	90	38	54

- Loops through columns 1 to 5.
- Within each column, loops through rows 1 to 10.
- Fills each cell with a random number between 0 and 100, rounded to the nearest whole number.

Offset

The Offset property in VBA is a way to refer to a cell that is a certain number of rows and columns away from a given cell or range.

```
Sub SimpleOffset()  
    ' Start at cell A1  
    Range("A1").Value = "Start"  
  
    ' Move 1 row down and 1 column to the right (cell B2)  
    Range("A1").Offset(1, 1).Value = "Moved"  
End Sub
```

- Range("A1").Value = "Start": Sets the value of cell A1 to "Start".
- Range("A1").Offset(1, 1).Value = "Moved": Moves the reference 1 row down and 1 column to the right (to cell B2) and sets its value to "Moved".

Do While Loop

The Do While loop in VBA is used to repeat a block of code as long as a specified condition is true.

```
Sub ExampleDoWhile()  
    Dim counter As Integer  
    counter = 1  
  
    Do While counter ≤ 5  
        MsgBox "Counter is: " & counter  
        counter = counter + 1  
    Loop  
End Sub
```

- counter = 1: Initializes a counter variable.
- Do While counter ≤ 5: Begins the loop with a condition that the loop will continue as long as counter is less than or equal to 5.
- MsgBox "Counter is: " & counter: Displays the current value of counter.
- counter = counter + 1: Increments the counter by 1.

- Loop: Ends the loop and returns to check the condition again.



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